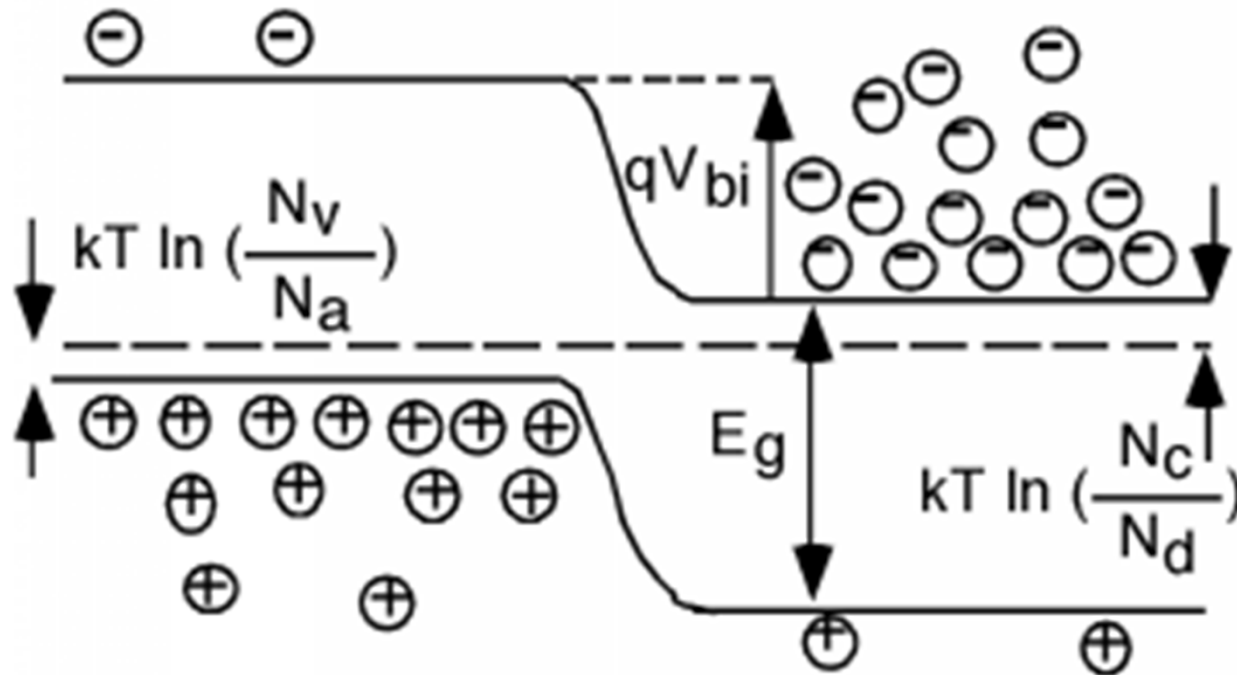


L7: The pn Junction



Dr. Shuming Chen

Department of Electrical and Electronic Engineering, SUSTech

Where do I see PN diodes everyday ?

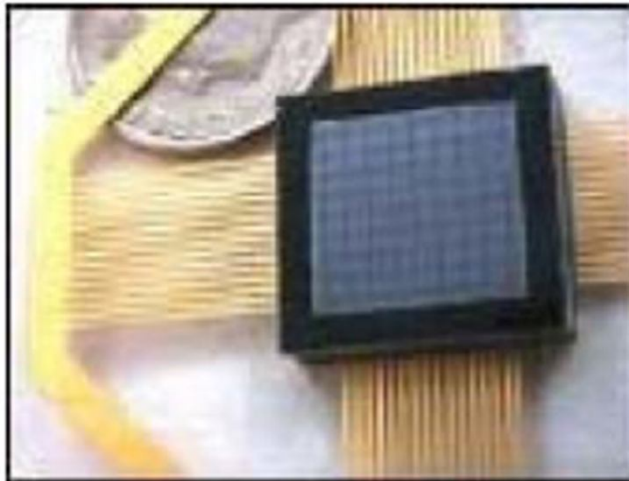
solar cells



laser



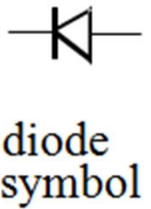
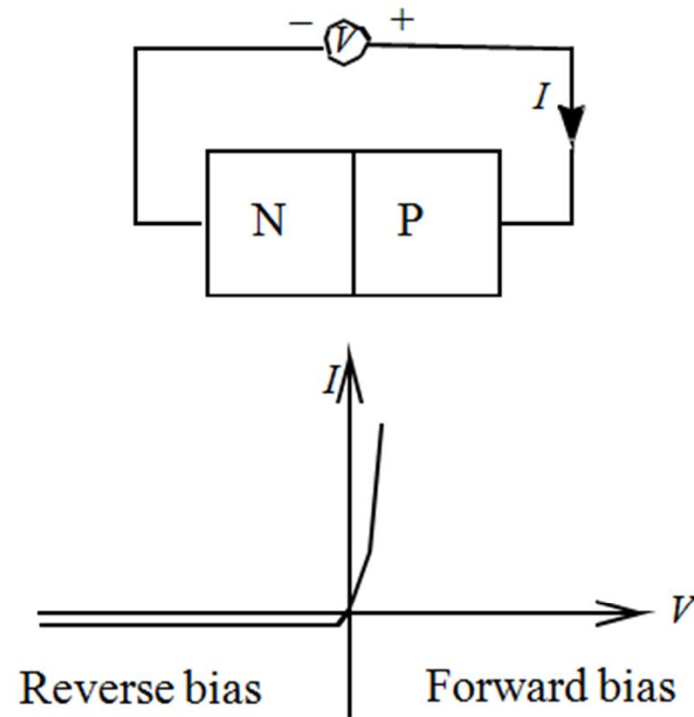
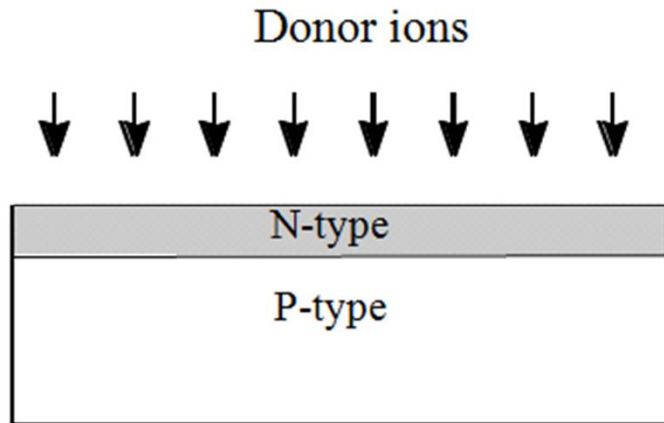
Avalanche Photodiode



OLED display



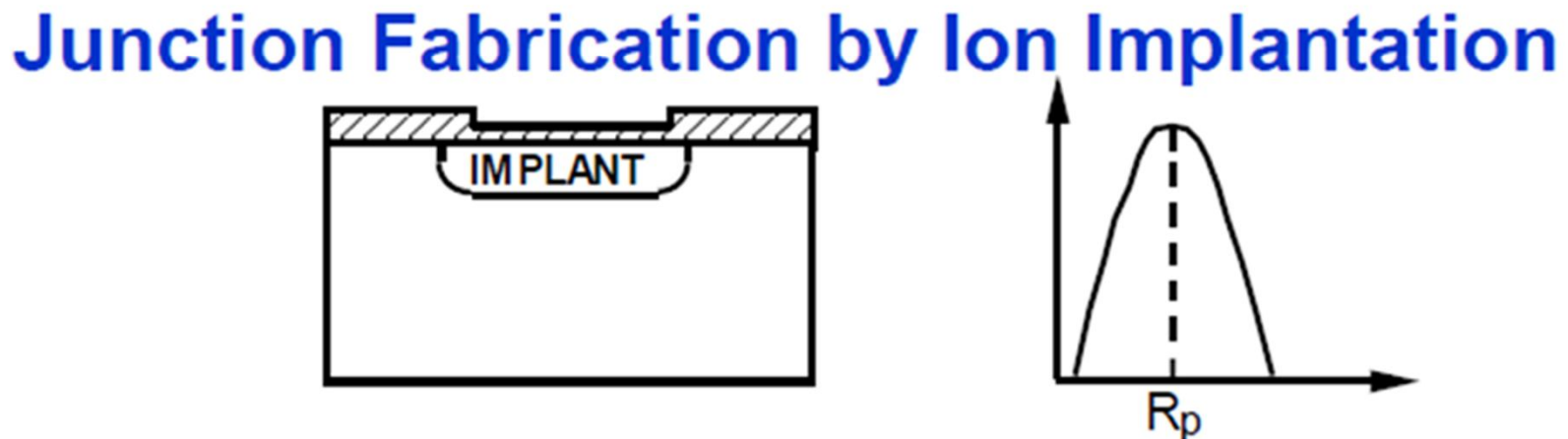
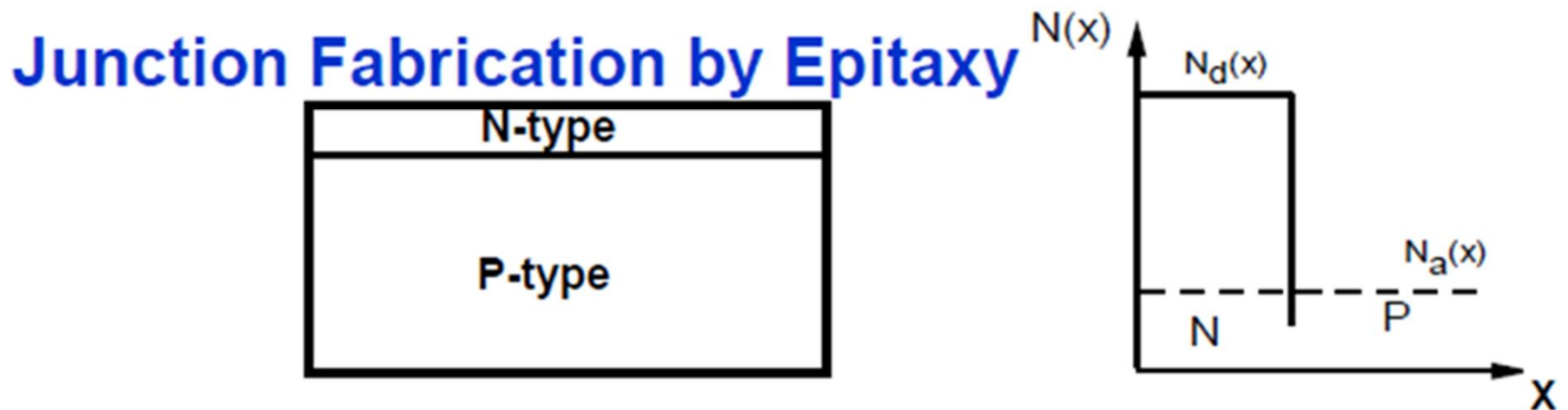
Building Blocks for semiconductor devices



PN junction is present in perhaps every semiconductor device.

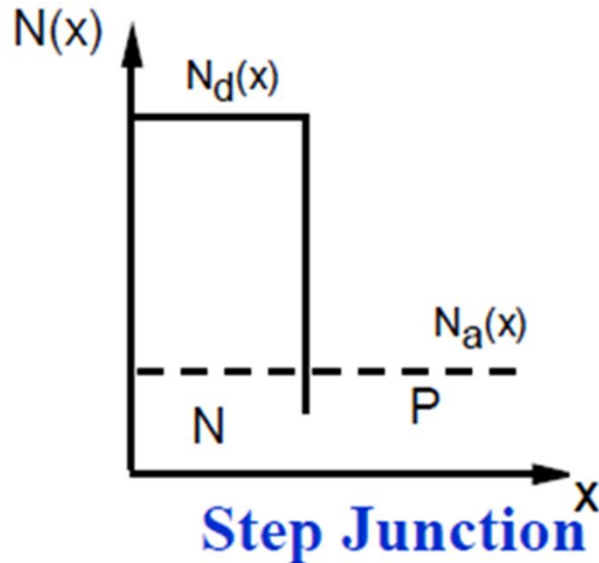
p-n Junction Formation

P-N junctions, in addition to being a useful semiconductor device, form the basis for almost all other semiconductor devices. Understanding their operation is basic to understanding most devices

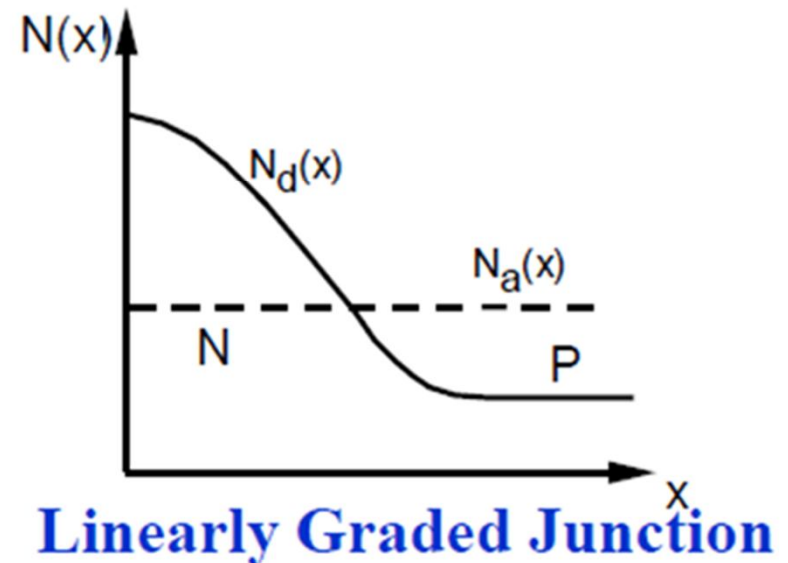


p-n Junction Profiles

The two profiles that are encountered most often in real devices are also easiest to analyze from a device physics point of view

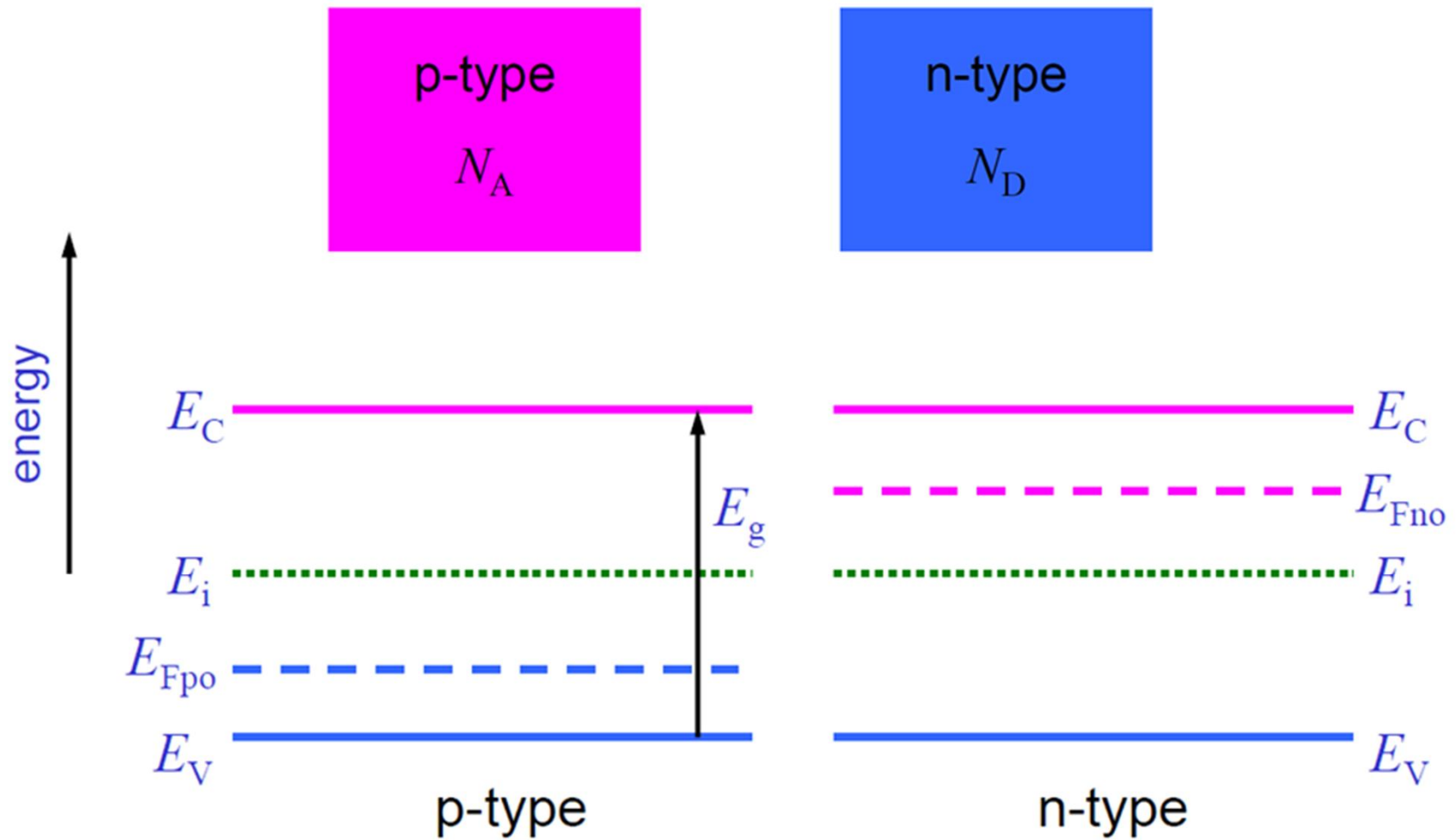


Good approximation for shallow, high concentration junctions ($x_j < 1\ \mu\text{m}$) and epitaxially grown junctions

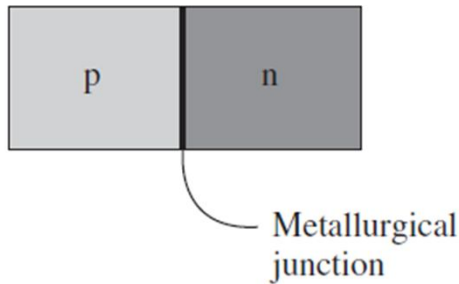


Good approximation for deep junctions ($x_j > 3\ \mu\text{m}$)

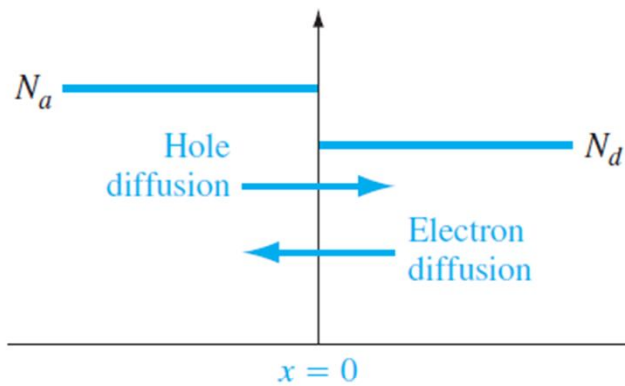
Before Connection



Connection

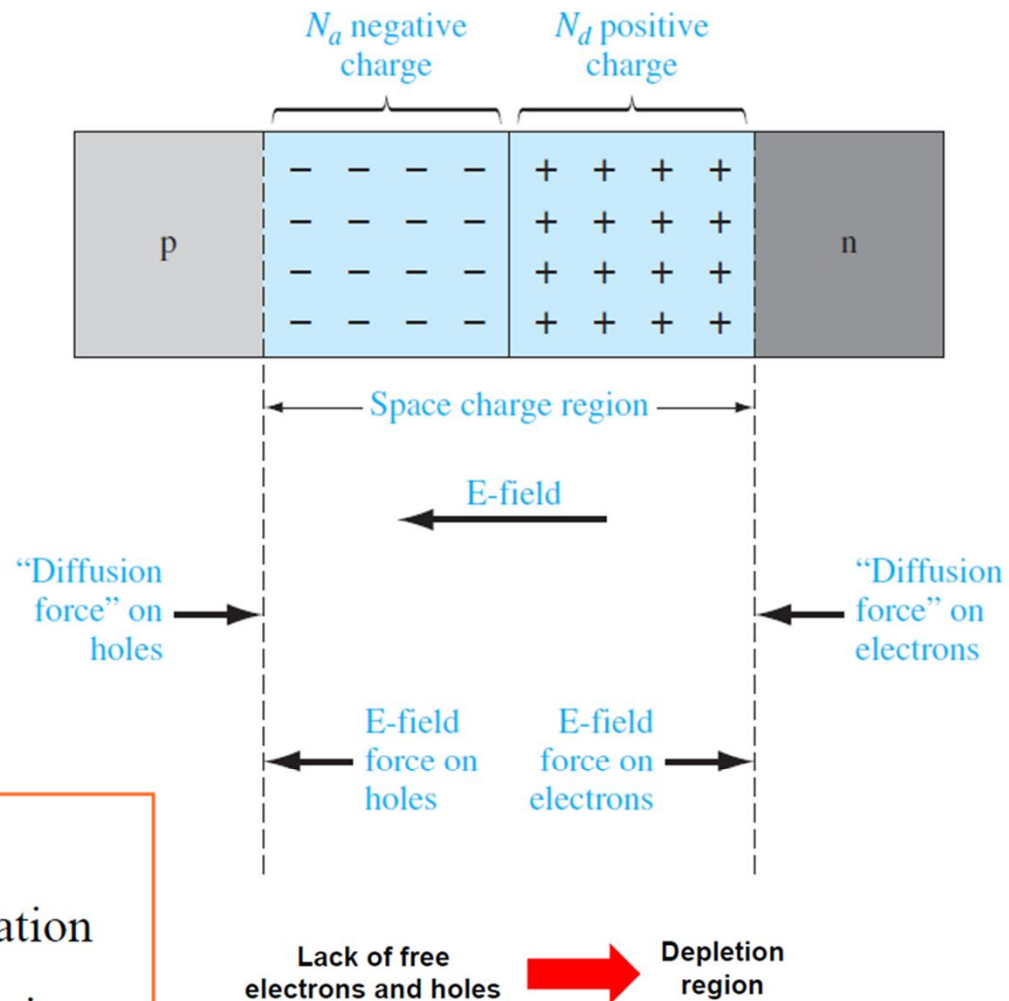


(a)



(b)

- No current
- No external excitation
- In thermal equilibrium
- Constant Fermi energy



Connection

Junction recombination:

✓ Some electrons in the n material diffuse across the junction into the p material and fill some of the holes.

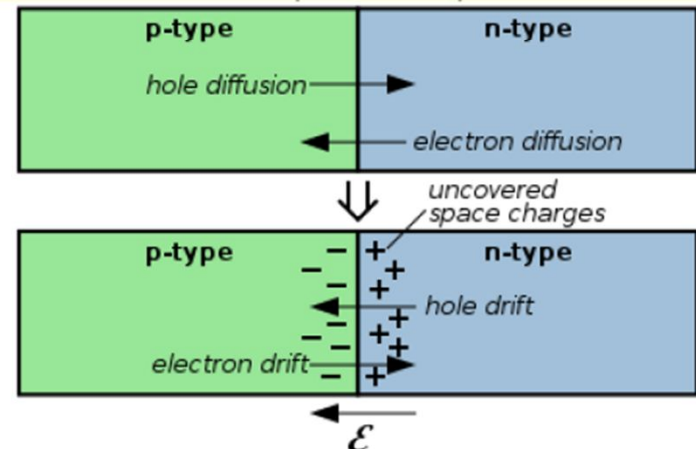
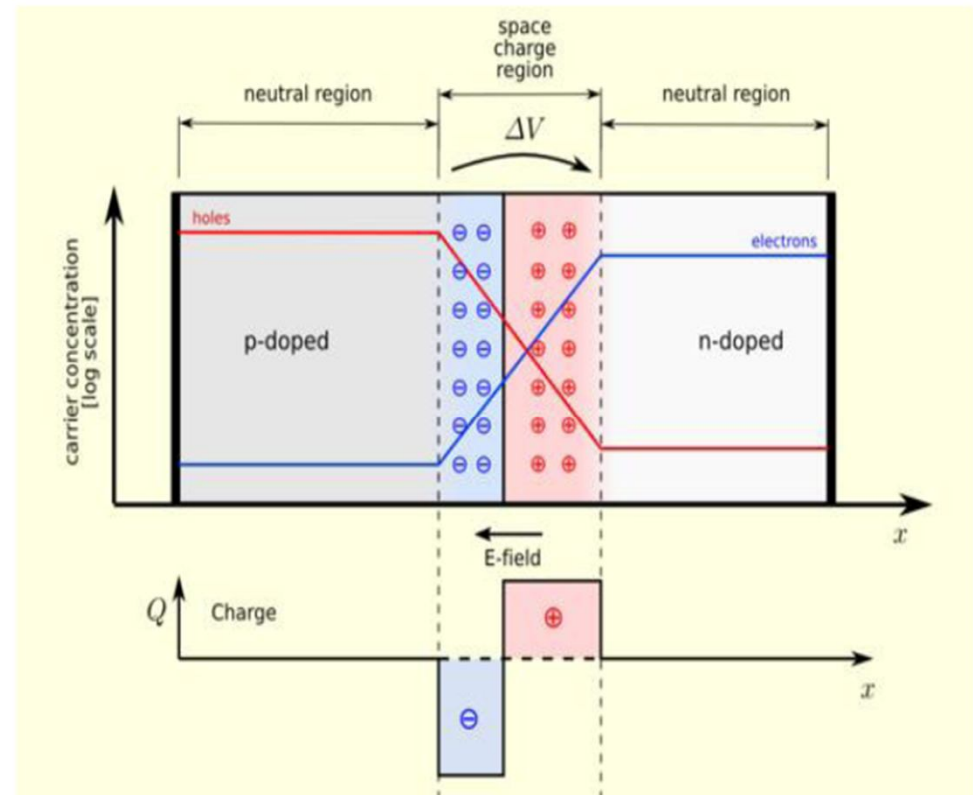
✓ Simultaneously, some holes in the p material diffuse across the junction into the n material and are filled by n material electrons.

Lack of free electrons and holes  Depletion region

Barrier potential :

✓ The flow of electrons and holes in opposite directions will create opposite charged ions on both sides establishing an electrostatic field.

✓ The diffusion will continue till the field is strong enough to prevent it.



Thermal equilibrium condition

At equilibrium condition, the drift current due to the electric field must exactly cancel the diffusion current due to the concentration gradient.

$$J_n = e\mu_n n \mathbf{E} + eD_n \frac{dn}{dx} = 0$$

$$J_p = e\mu_p p \mathbf{E} - eD_p \frac{dp}{dx} = 0$$

$$\therefore \frac{dE_F}{dx} = 0$$

the Fermi levels in the p- and n-type semiconductors must be equal

Thermal equilibrium condition

When a semiconductor is in equilibrium, there's no *net* flux

$$\text{Since } F_n = D_n \frac{dn}{dx} + \mu_n n \mathcal{E} = 0$$

$$n = n_i e^{(E_F - E_i)/kT}$$

$$\frac{dn}{dx} = \frac{n}{kT} \left[\frac{dE_F}{dx} - \frac{dE_i}{dx} \right]$$

Then

$$F_n = \frac{D_n}{kT} n \left[\frac{dE_F}{dx} - \frac{dE_i}{dx} \right] + \mu_n n \frac{1}{q} \frac{dE_i}{dx}$$

Using the Einstein relationship,

$$\frac{D_n}{kT} = \frac{\mu_n}{q}$$

$$F_n = \frac{D_n}{kT} n \frac{dE_F}{dx} \quad \text{A gradient in the Fermi level is the driving force for carrier motion}$$

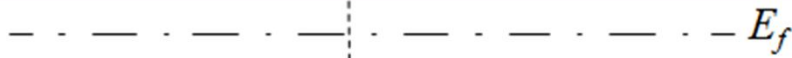
In equilibrium, $F_n = 0 \Rightarrow E_F = \text{constant.}$

This is an extremely important result!

Energy Band Diagram of a PN Junction

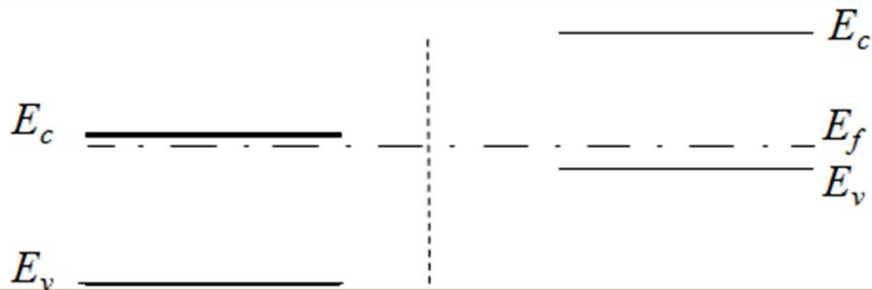
N-region ← → **P-region**

(a)



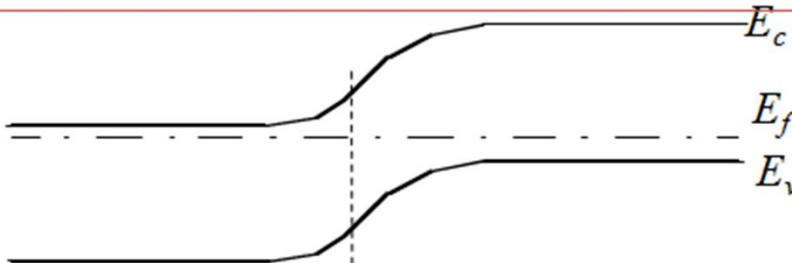
E_f is constant at equilibrium

(b)



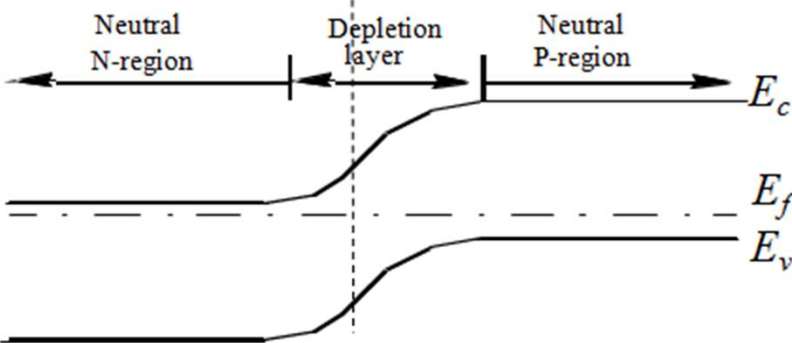
E_c and E_v are known relative to E_f

(c)



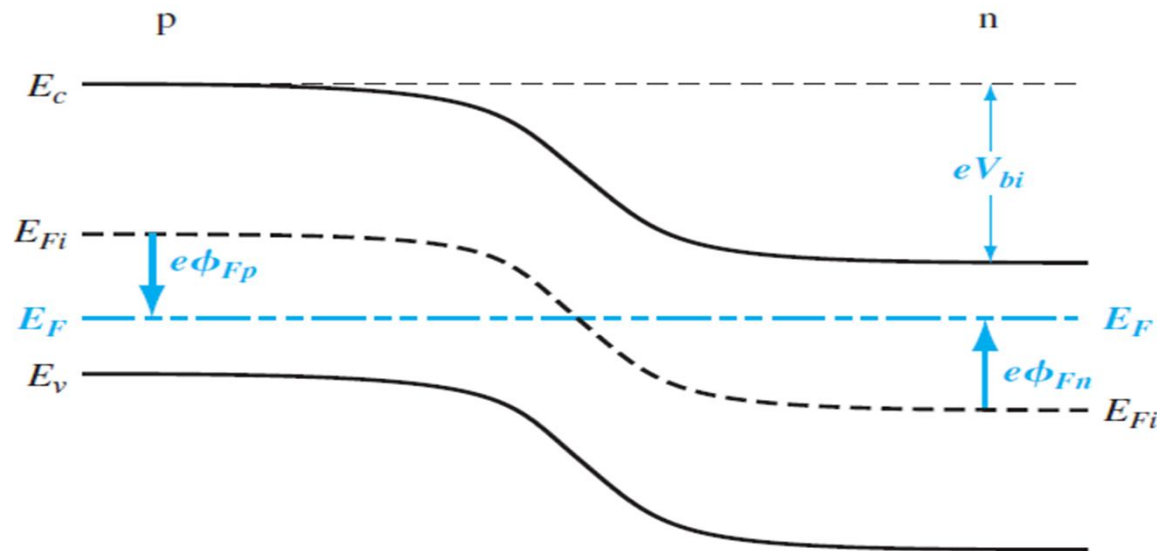
E_c and E_v are smooth, the exact shape to be determined.

(d)



A depletion layer exists at the PN junction where $n \approx 0$ and $p \approx 0$.

7.2.1 Built-in Potential Barrier



Thus, with constant Fermi level,

$$V_{bi} = \phi_{Fn} - \phi_{Fp} = \text{built in potential}$$

$$= \frac{kT}{q} \left[\ln \frac{N_D}{n_i} + \ln \frac{N_A}{n_i} \right]$$

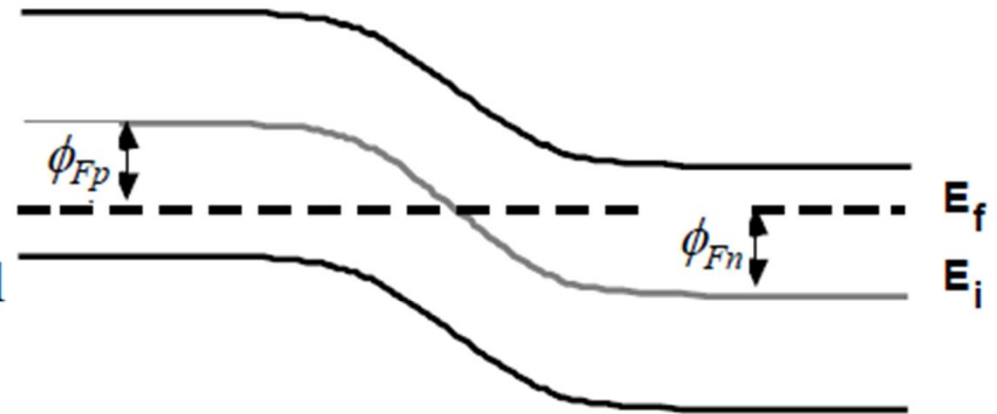
$$= \frac{kT}{q} \ln \frac{N_D N_A}{n_i^2}$$

An important point: previously N_a and N_d denoted the concentrations in the same region. From now on, they will denote the NET concentrations in the individual p- and n-regions, respectively.

The built in potential is only dependent upon the carrier densities, not diffusion coefficient, lifetime, etc. The higher the doping, the larger the built in potential.

Built-in Potential Barrier

V_{bi} : Quiz



$$V_{bi} = \phi_{Fn} - \phi_{Fp} = \text{built in potential}$$

$$= \frac{kT}{q} \left[\ln \frac{N_D}{n_i} + \ln \frac{N_A}{n_i} \right]$$

$$= \frac{kT}{q} \ln \frac{N_D N_A}{n_i^2}$$

- What is the maximum value of V_{bi} ?
- Does V_{bi} increase / decrease with temperature ?

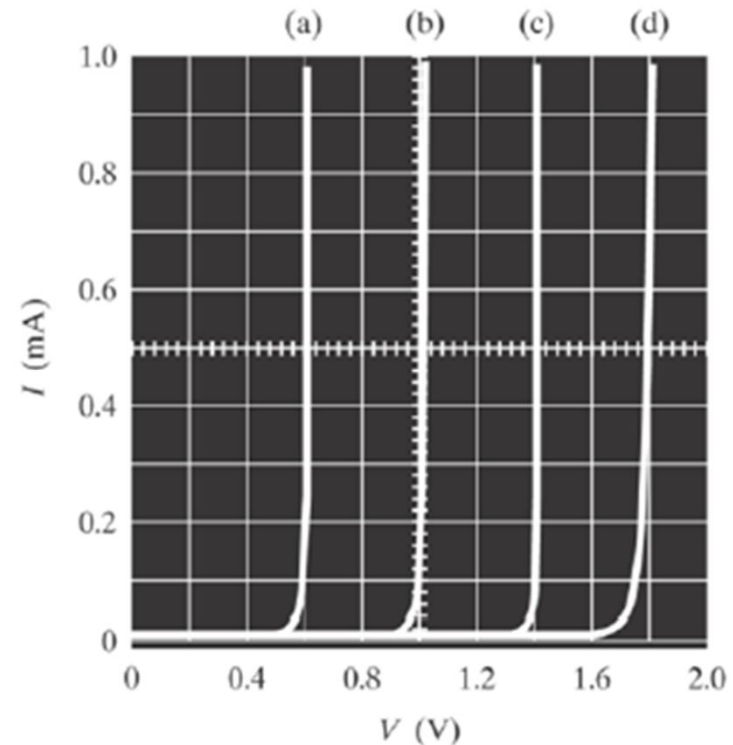
Built-in Potential Barrier

$$V_{bi} = \frac{kt}{q} \ln \frac{N_d N_a}{n_i^2} = \frac{kt}{q} \ln \frac{N_d N_a}{N_c N_v \exp(-\frac{E_g}{kt})} \approx \frac{E_g}{q}, \text{ for heavy doping}$$

Hole injection and therefore the current is small if the forward bias V is much less than the V_0 (or E_{gap} for $p^+ - n^+$ junctions).

- a) Ge $E_g \approx 0.7 \text{ eV}$
- b) Si $E_g \approx 1.1 \text{ eV}$
- c) GaAs $E_g \approx 1.4 \text{ eV}$
- d) GaAsP $E_g \approx 1.9 \text{ eV}$

In most cases, the built-in voltage is approximately 80% of the material bandgap



Poisson's equation

$$\begin{aligned}\frac{d^2\psi(x)}{dx^2} &= -\frac{d\mathbf{E}(x)}{dx} = -\frac{\rho_s(x)}{\varepsilon} = \\ &= -\frac{e}{\varepsilon} [N_D(x) - N_A(x) + p(x) - n(x)]\end{aligned}$$

ψ - electrostatical potential

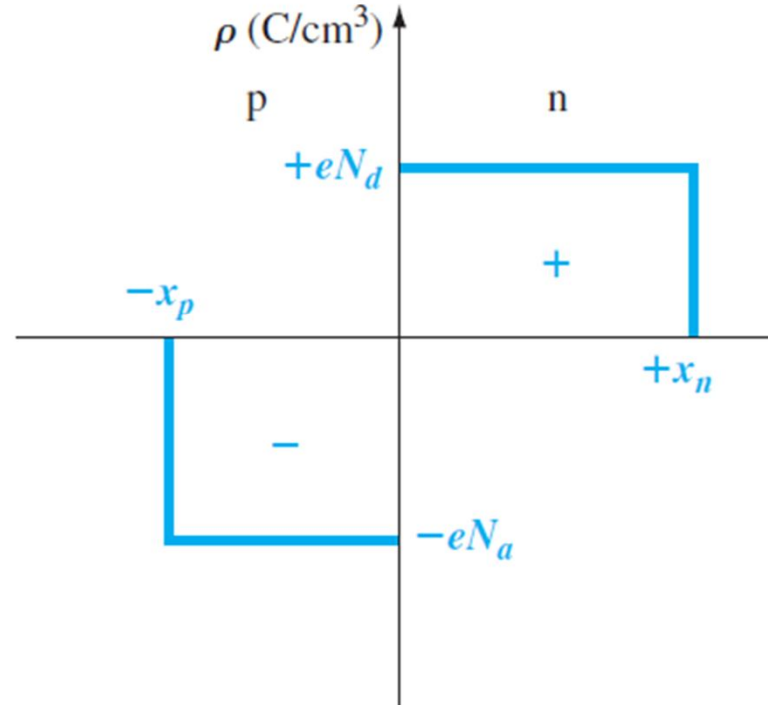
ρ - space charge density

ε - semiconductor permittivity

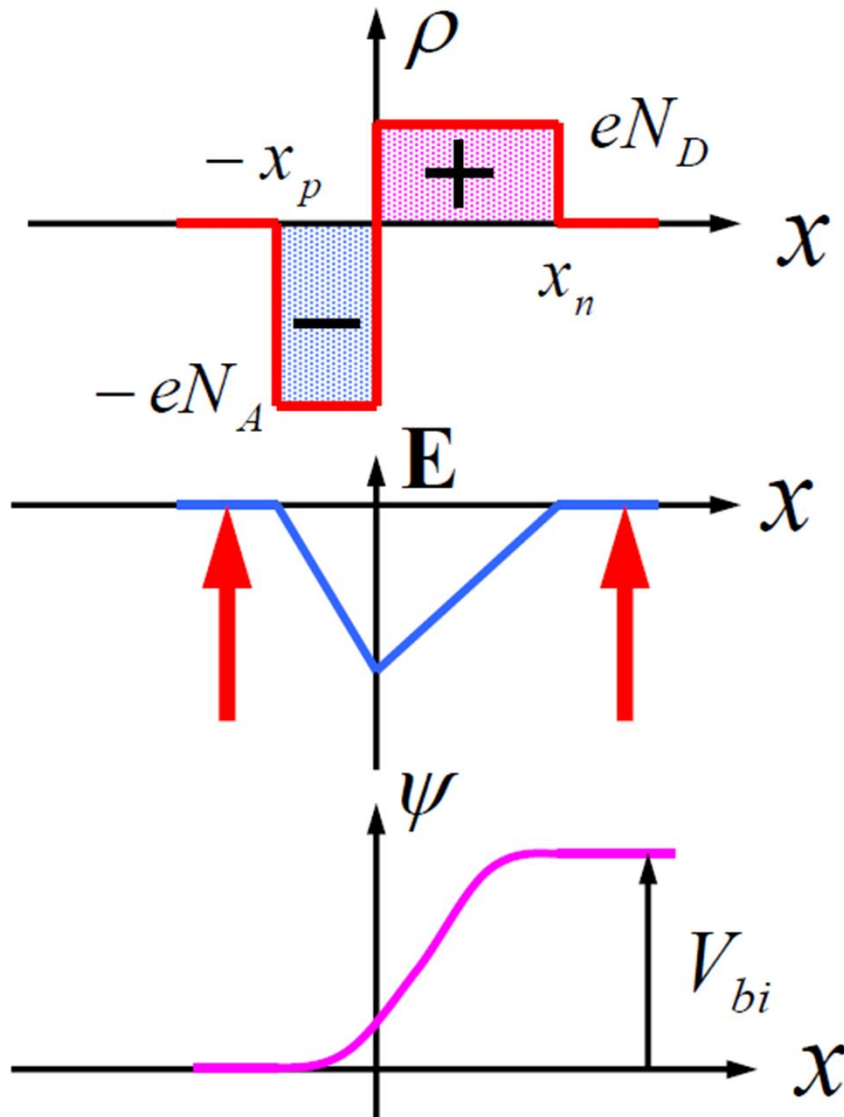
Poisson's equation for abrupt junction

$$\frac{d^2\psi(x)}{dx^2} = -\frac{d\mathbf{E}(x)}{dx} = +\frac{eN_A}{\varepsilon} \quad \text{for} \quad -x_p \leq x < 0$$

$$\frac{d^2\psi(x)}{dx^2} = -\frac{d\mathbf{E}(x)}{dx} = -\frac{eN_D}{\varepsilon} \quad \text{for} \quad 0 < x \leq x_n$$

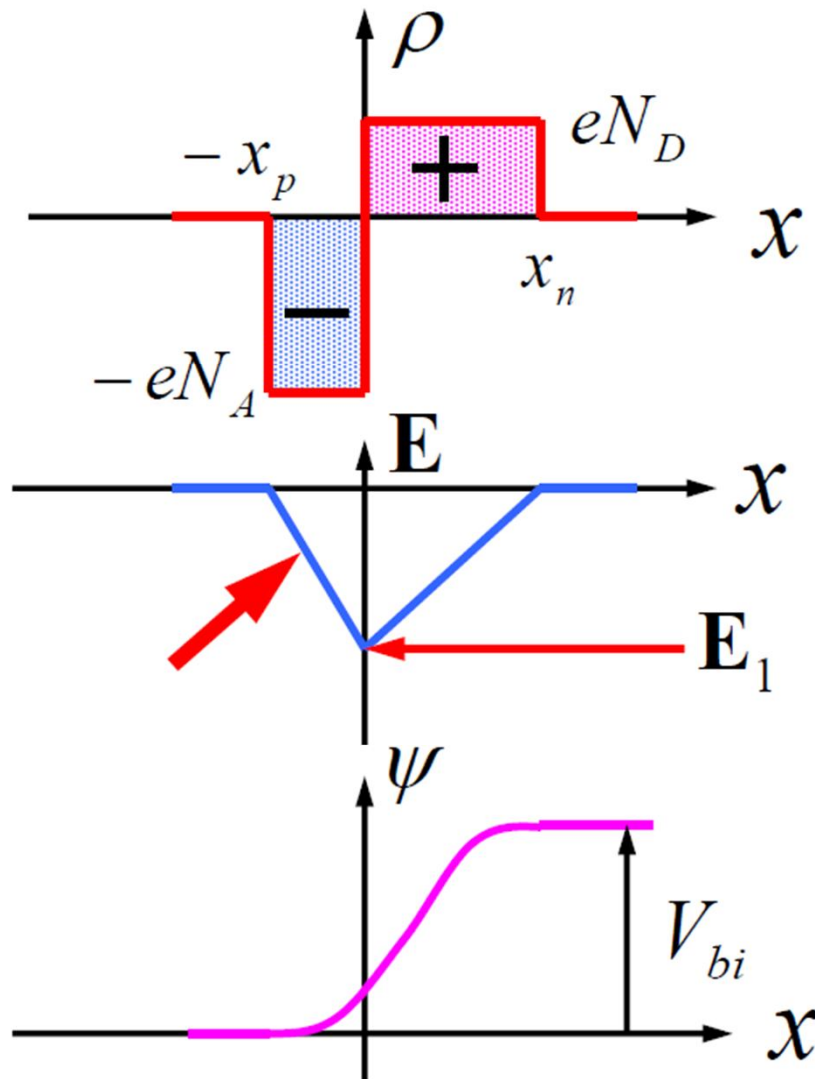


7.2.2 Electric Field



$$\frac{d\mathbf{E}(x)}{dx} = 0$$

Electric Field Distribution



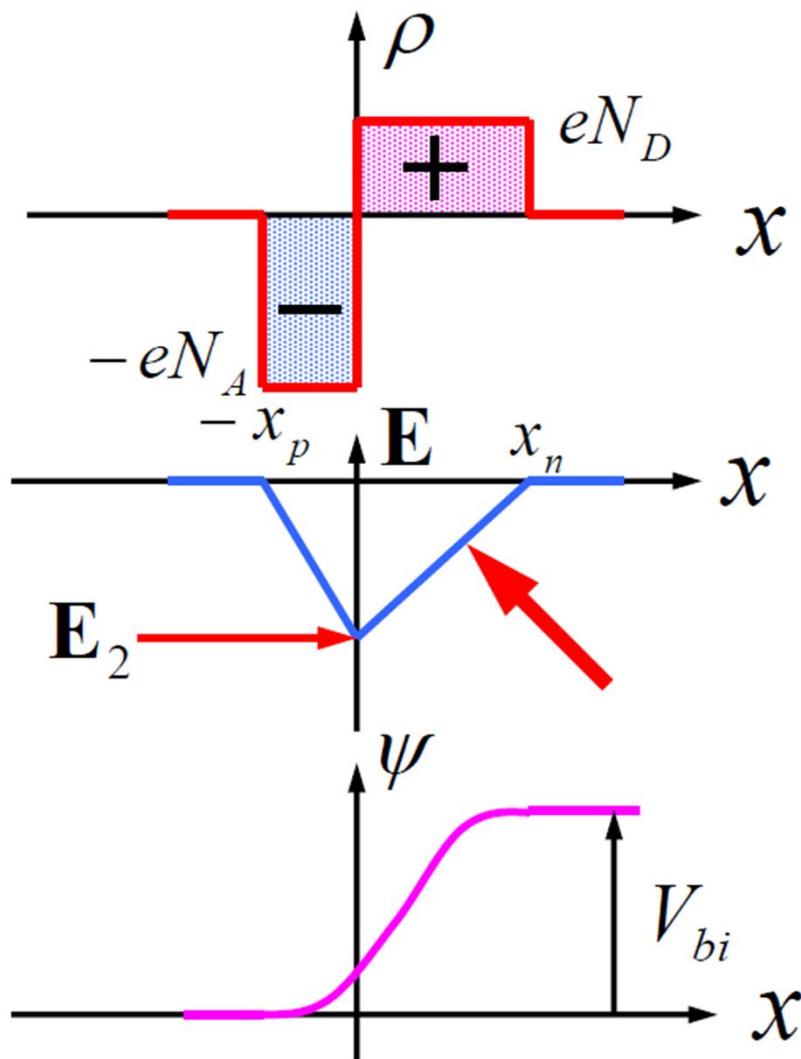
$$\frac{d\mathbf{E}(x)}{dx} = \frac{-eN_A}{\epsilon}$$

$$\mathbf{E}(x) = -\frac{eN_A}{\epsilon}x + \mathbf{E}_1$$

$$\mathbf{E}_1 = -\frac{eN_A}{\epsilon}x_p$$

$$\mathbf{E}(x) = -\frac{eN_A}{\epsilon}(x + x_p)$$

Electric Field Distribution



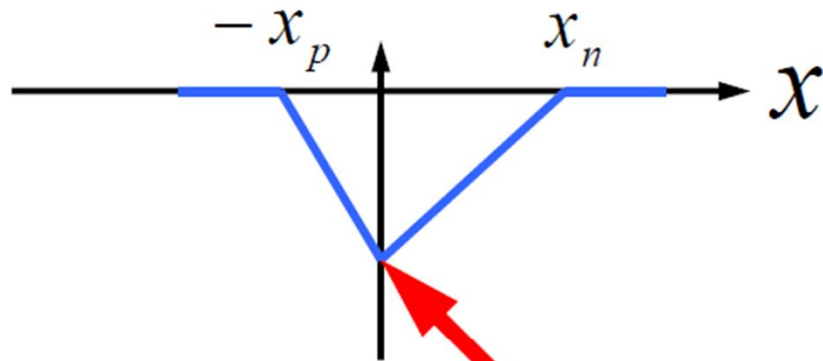
$$\frac{d\mathbf{E}(x)}{dx} = \frac{eN_D}{\epsilon}$$

$$\mathbf{E}(x) = \frac{eN_D}{\epsilon} x + \mathbf{E}_2$$

$$\mathbf{E}_2 = -\frac{eN_D}{\epsilon} x_n$$

$$\mathbf{E}(x) = \frac{eN_D}{\epsilon} (x - x_n)$$

Maximum electric field



$$\mathbf{E}_{\max} = \mathbf{E}(0) = -\frac{eN_A}{\epsilon} x_p = -\frac{eN_D}{\epsilon} x_n$$

consequence:

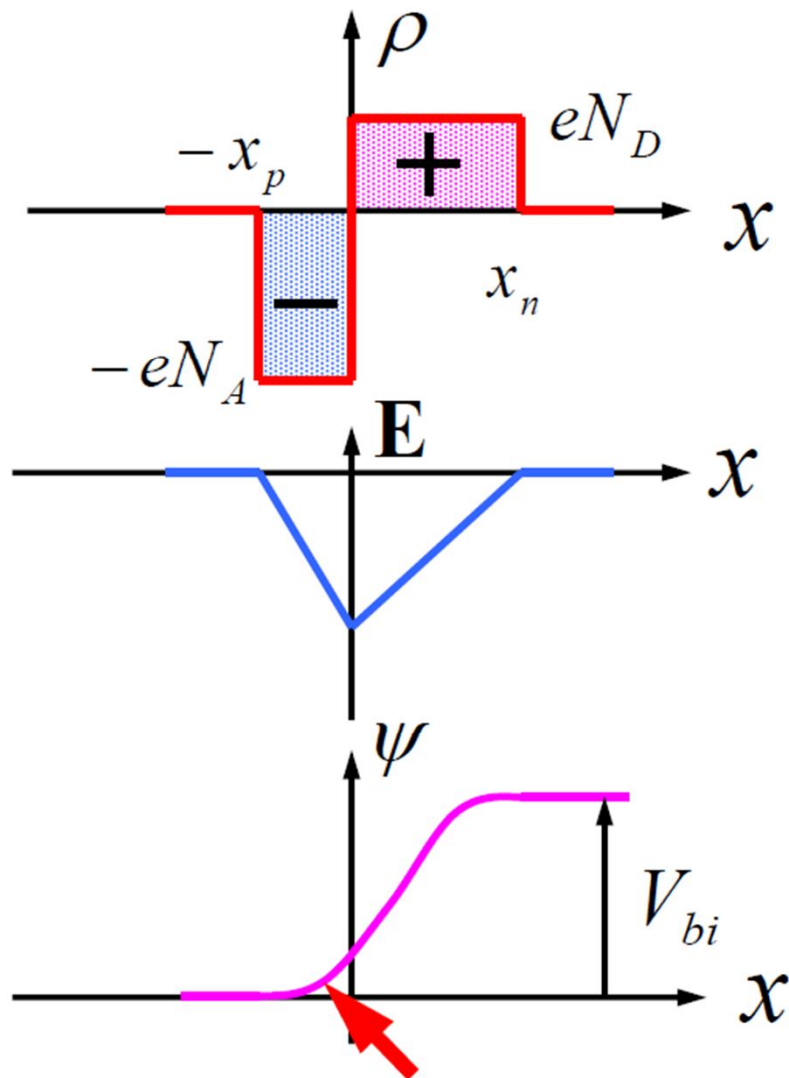
$$N_A x_p = N_D x_n$$

Potential distribution

$$\mathbf{E}(x) = -\frac{d\psi(s)}{dx}$$

$$\psi(s) = -\int \mathbf{E}(x)dx$$

Potential distribution



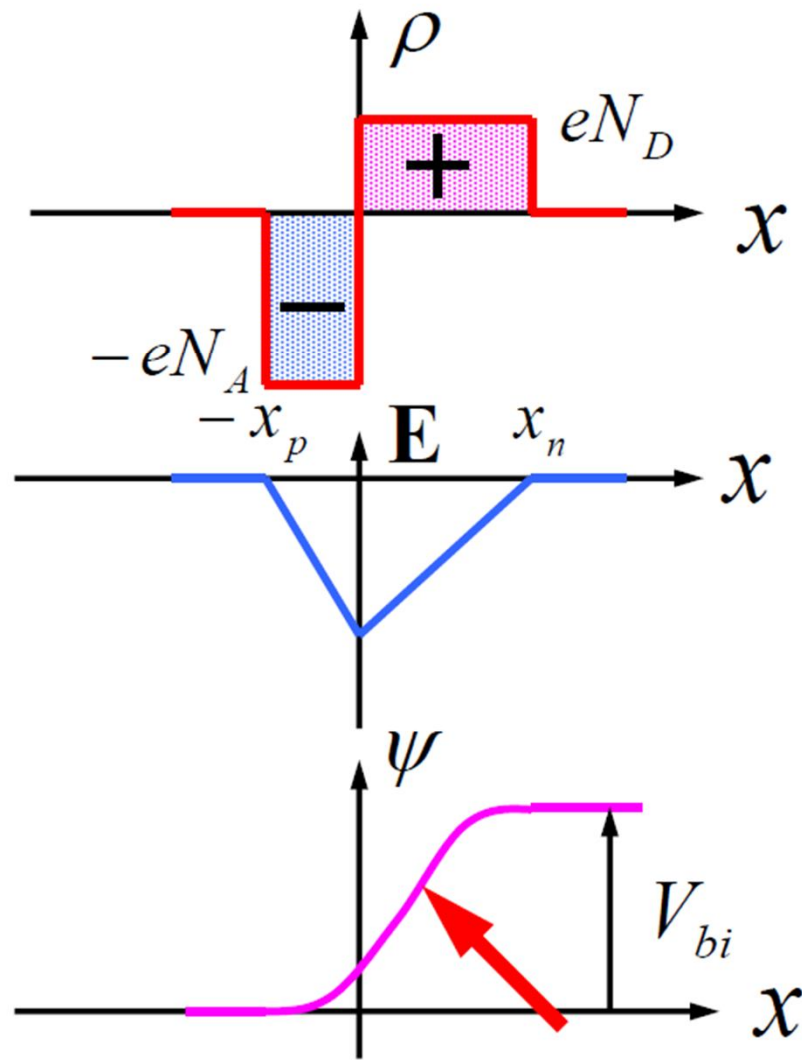
$$\begin{aligned}\psi(x) &= \int \frac{eN_A}{\epsilon} (x + x_p) dx \\ &= \frac{eN_A}{\epsilon} \left(\frac{x^2}{2} + x_p x \right) + \psi_1\end{aligned}$$

with $\psi(-x_p) = 0$

$$\psi_1 = \frac{eN_A}{\epsilon} \frac{x_p^2}{2}$$

$$\psi(x) = \frac{eN_A}{2\epsilon} (x_p + x)^2$$

Potential distribution



$$\begin{aligned}\psi(x) &= \int \frac{eN_D}{\varepsilon} (x_n + x) dx \\ &= \frac{eN_D}{\varepsilon} \left(x_n x - \frac{x^2}{2} \right) + \psi_2\end{aligned}$$

with $\psi(x_n) = V_{bi}$

$$\psi_2 = V_{bi} - \frac{eN_D}{\varepsilon} \frac{x_n^2}{2}$$

$$\psi(x) = V_{bi} - \frac{eN_D}{2\varepsilon} (x_n - x)^2$$

Potential distribution

for $x = 0$ both expressions

$$\psi(x) = V_{bi} - \frac{eN_D}{2\varepsilon} (x_n - x)^2$$

$$\psi(x) = \frac{eN_A}{2\varepsilon} (x_p + x)^2$$

must give the same value:

$$\psi(0) = V_{bi} - \frac{eN_D}{2\varepsilon} x_n^2 = \frac{eN_A}{2\varepsilon} x_p^2$$

$$V_{bi} = \frac{e}{2\varepsilon} (N_D x_n^2 + N_A x_p^2)$$

Depletion width

$$x_d \equiv x_n + x_p$$

$$V_{bi} = \frac{e}{2\varepsilon} (N_D x_n^2 + N_A x_p^2) \quad N_A x_p = N_D x_n$$

$$V_{bi} = \frac{e}{2\varepsilon} \left[N_D x_n^2 + N_A \left(\frac{N_D x_n}{N_A} \right)^2 \right] = x_n^2 \frac{e}{2\varepsilon} \left(N_D + \frac{N_D^2}{N_A} \right)$$

$$V_{bi} = \frac{e}{2\varepsilon} \left[N_D \left(\frac{N_A x_p}{N_D} \right)^2 + N_A x_p^2 \right] = x_p^2 \frac{e}{2\varepsilon} \left(\frac{N_A^2}{N_D} + N_A \right)$$

$$x_n = \sqrt{\frac{2\varepsilon}{e} \frac{N_A}{N_D N_A + N_D^2} V_{bi}}$$

$$x_p = \sqrt{\frac{2\varepsilon}{e} \frac{N_D}{N_D N_A + N_A^2} V_{bi}}$$

Depletion width

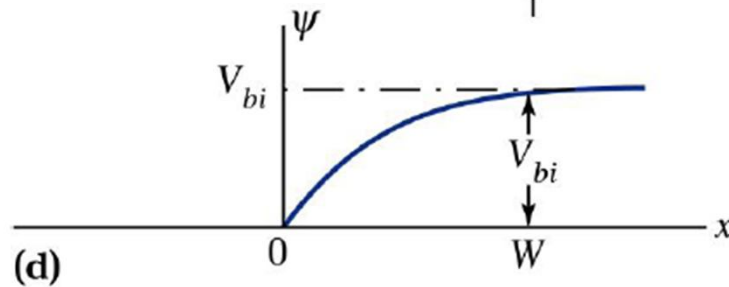
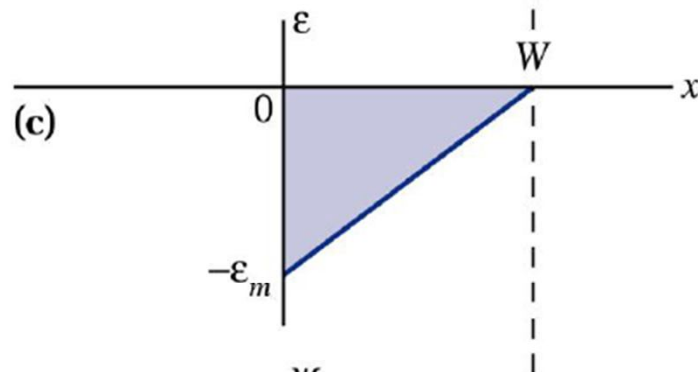
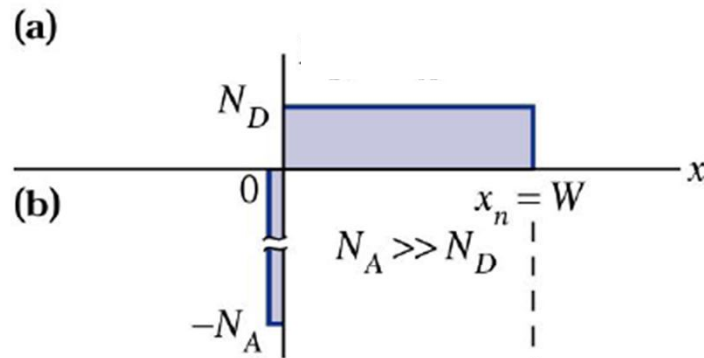
$$x_n = \sqrt{\frac{2\varepsilon}{e} \frac{N_A}{N_D N_A + N_D^2} V_{bi}}$$

$$x_p = \sqrt{\frac{2\varepsilon}{e} \frac{N_D}{N_D N_A + N_A^2} V_{bi}}$$

$$\begin{aligned} x_d^2 &= (x_n + x_p)^2 = \frac{2\varepsilon}{e} \frac{N_A}{N_D N_A + N_D^2} V_{bi} + \frac{2\varepsilon}{e} \frac{N_D}{N_A N_D + N_A^2} V_{bi} \\ &+ 2 \cdot \sqrt{\frac{2\varepsilon}{e} \frac{N_A}{N_D N_A + N_D^2} V_{bi}} \cdot \sqrt{\frac{2\varepsilon}{e} \frac{N_D}{N_A N_D + N_A^2} V_{bi}} = \\ &\frac{2\varepsilon}{e} V_{bi} \frac{N_A^2 + 2N_D N_A + N_D^2}{(N_A + N_D) N_D N_A} = \frac{2\varepsilon (N_A + N_D)}{e N_D N_A} V_{bi} \end{aligned}$$

$$x_d = \sqrt{\frac{2\varepsilon (N_A + N_D)}{e N_D N_A} V_{bi}}$$

One-side abrupt junction

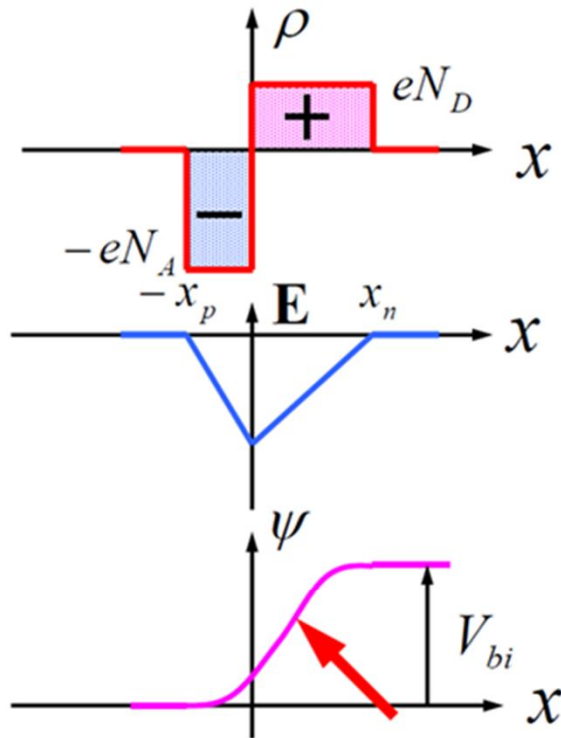


$$x_d = \sqrt{\frac{2\varepsilon}{e} \frac{(N_A + N_D)}{N_D N_A} V_{bi}}$$

if $N_A \gg N_D$

$$x_d \cong x_n = \sqrt{\frac{2\varepsilon V_{bi}}{e N_D}}$$

pn junction summary



$$1. \quad V_{bi} = \frac{kT}{q} \ln \frac{N_D N_A}{n_i^2}$$

2.

$$x_n = \sqrt{\frac{2\epsilon}{e} \frac{N_A}{N_D N_A + N_D^2} V_{bi}}$$

$$x_p = \sqrt{\frac{2\epsilon}{e} \frac{N_D}{N_D N_A + N_A^2} V_{bi}}$$

$$x_d = \sqrt{\frac{2\epsilon}{e} \frac{(N_A + N_D)}{N_D N_A} V_{bi}}$$

3.

$$E(x) = -\frac{eN_A}{\epsilon} (x + x_p)$$

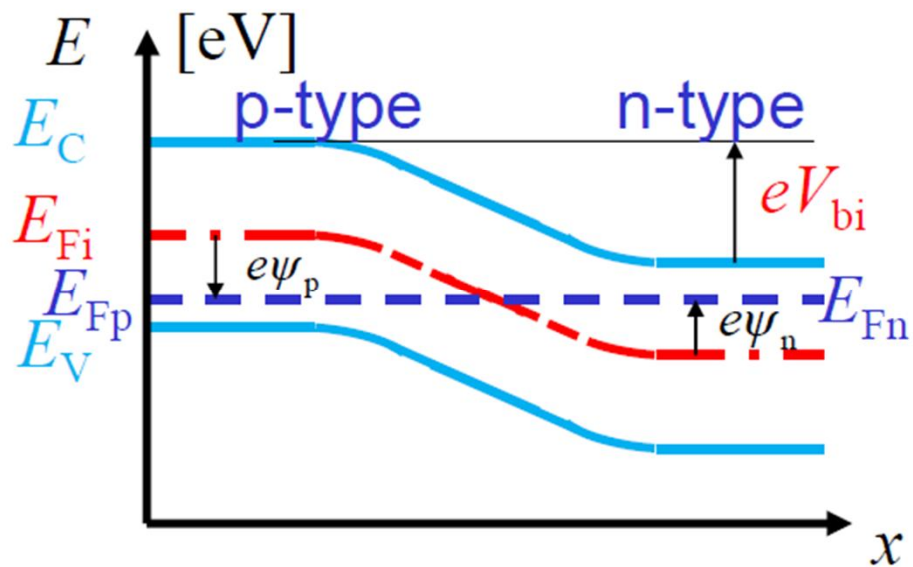
$$E(x) = \frac{eN_D}{\epsilon} (x - x_n)$$

4.

$$\psi(x) = \frac{eN_A}{2\epsilon} (x_p + x)^2$$

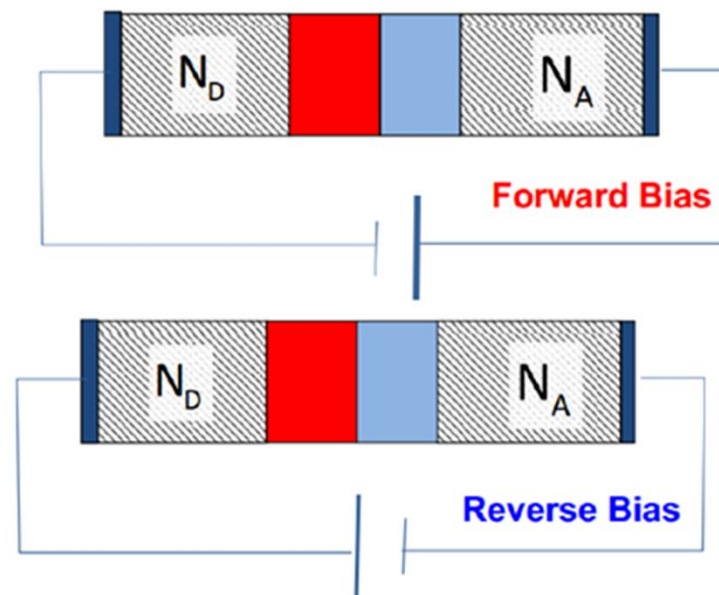
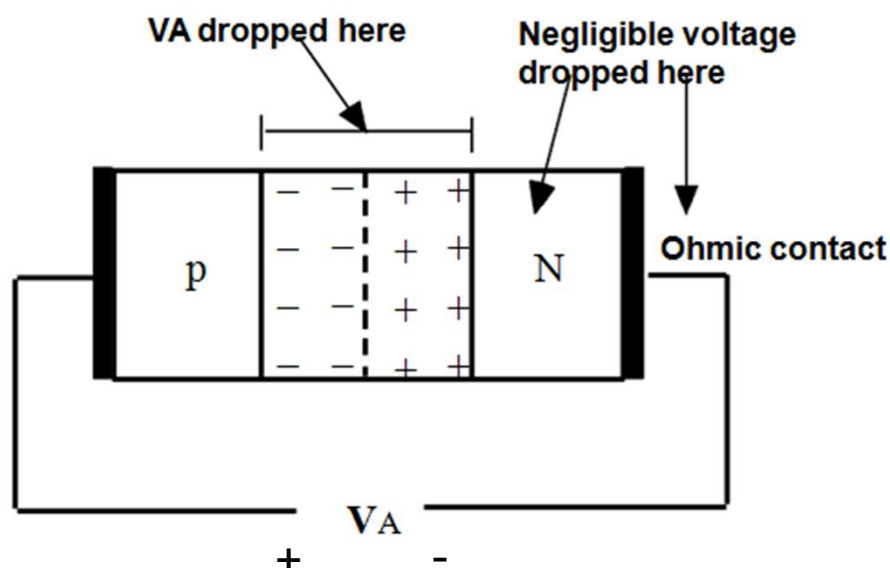
$$\psi(x) = V_{bi} - \frac{eN_D}{2\epsilon} (x_n - x)^2$$

Unbiased junction



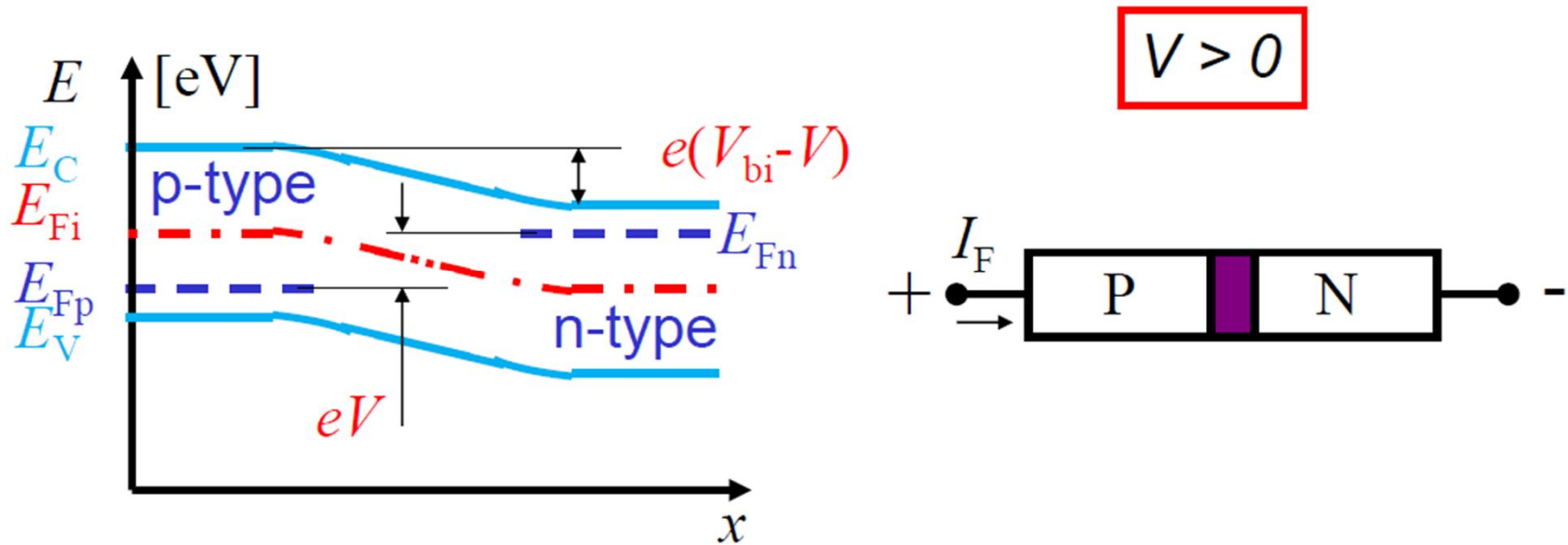
Biased junction

Quiz: where does the voltage drop occur ?



Applied voltage appears essentially entirely across the depletion region. The total potential drop across the junction is , $V_{bi} - V_A$.

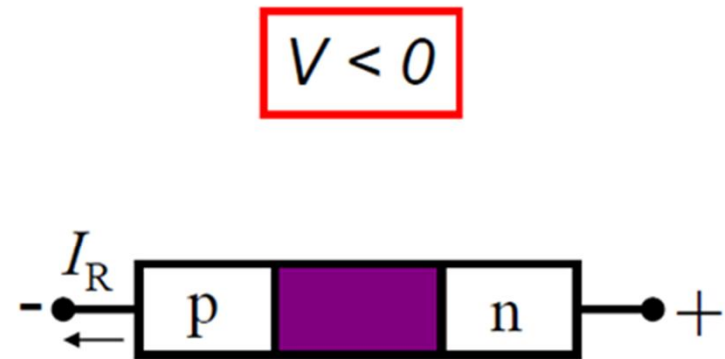
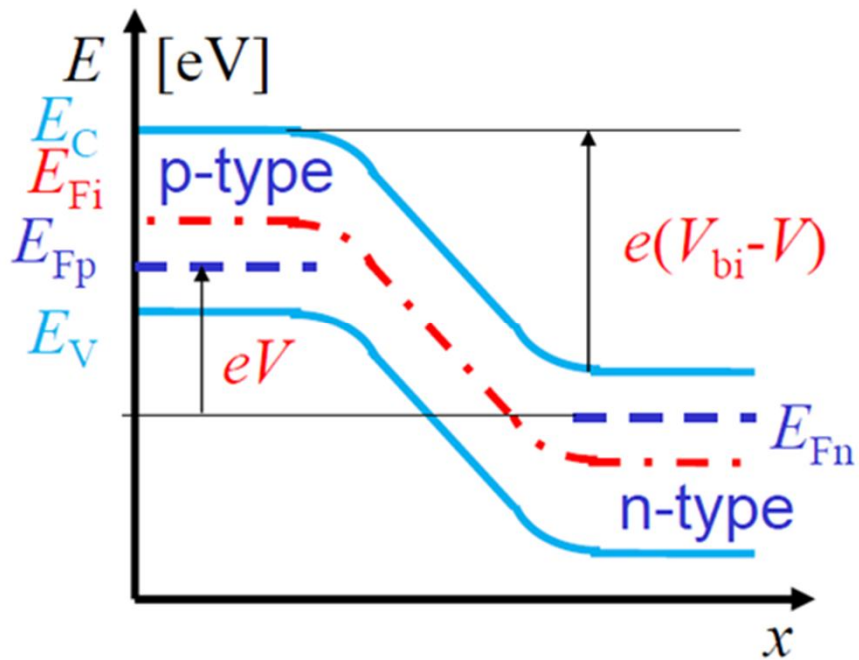
Forward-biased junction



E -field across depletion region decreases.

W decreases, diffusion pushes mobile carriers towards the junction.

Reverse-biased junction



Depletion layer width for polarized junction

$$x_d = \sqrt{\frac{2\varepsilon}{e} \frac{(N_A + N_D)}{N_D N_A} (V_{bi} - V)}$$

convention: V - is positive for positive on p and negative on n

for one side abrupt junction:

$$x_d = \sqrt{\frac{2\varepsilon(V_{bi} - V)}{eN_B}}$$

N_B – lightly doped bulk concentration

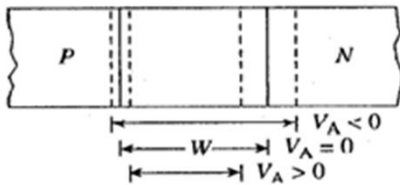
7.3.2 Junction Capacitance

The junction depletion layer capacitance per unit area is defined as:

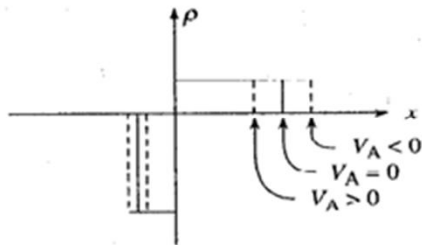
$$C_j = \frac{dQ}{dV}$$

Where dQ is the incremental charge in depletion layer charge per unit area for an incremental change in the applied voltage dV .

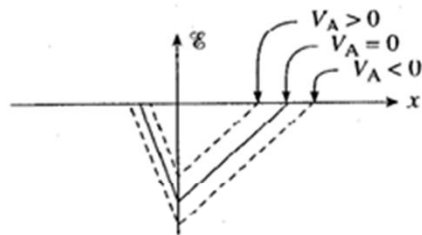
pn junction capacitance



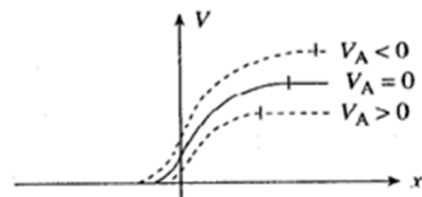
(a)



(b)



(c)



(d)

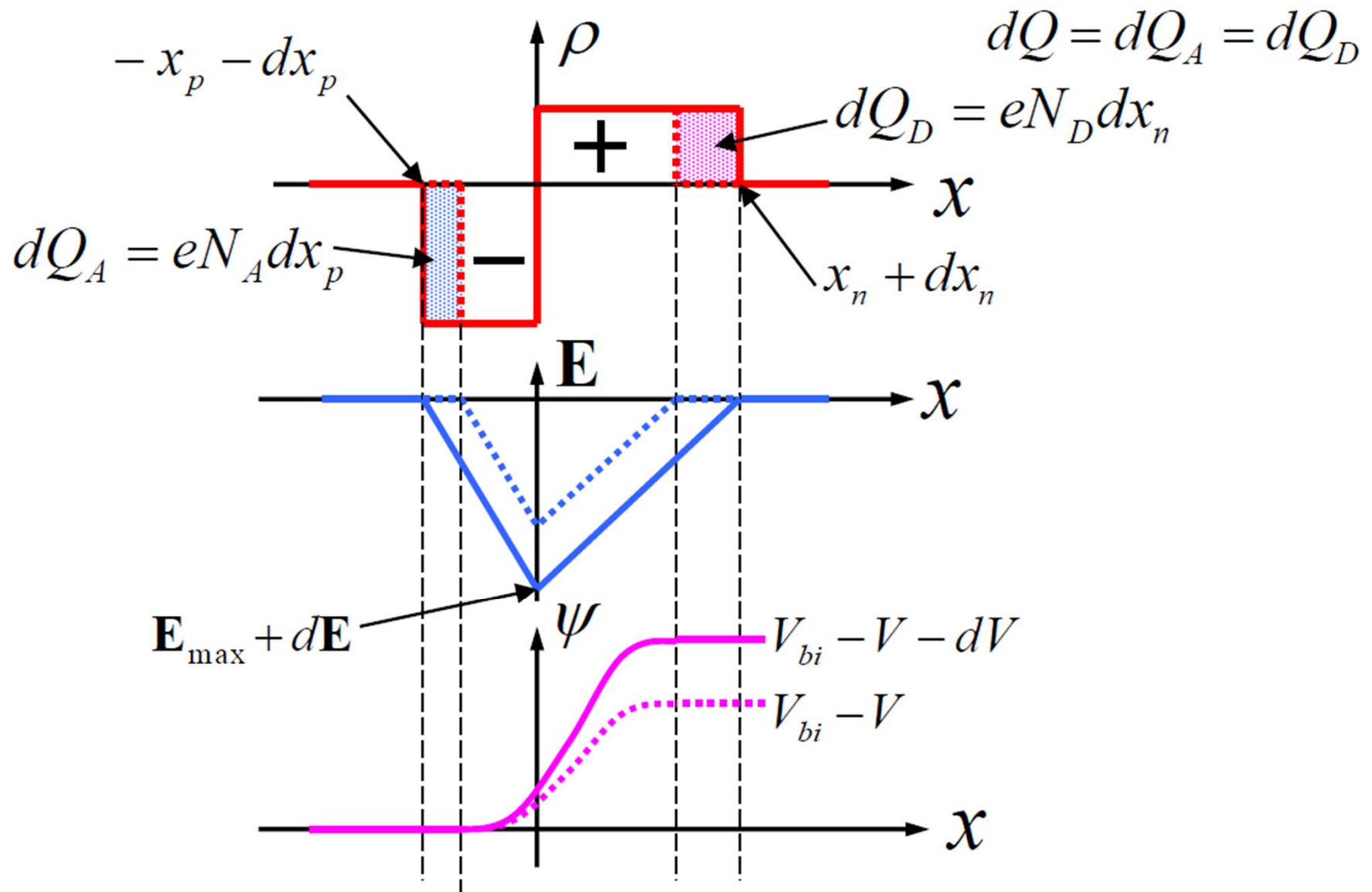
The small signal capacitance of the junction is given by

$$C = \frac{dQ}{dV_A} = \frac{AK_s \epsilon_o}{W} \quad (16)$$

where A = cross-sectional area

This result holds for any arbitrary doping profile. Given any profile, if W is determined, the small signal capacitance C can be calculated. W depends on the dopant profile and the applied bias

Junction capacitance



Capacitance of abrupt junction

$$C_j = \frac{dQ}{dV} \qquad dQ = eN_D dx_n = eN_A dx_p$$

$$x_n = \sqrt{\frac{2\varepsilon}{q} \frac{N_A}{N_D N_A + N_D} (V_{bi} - V)}$$

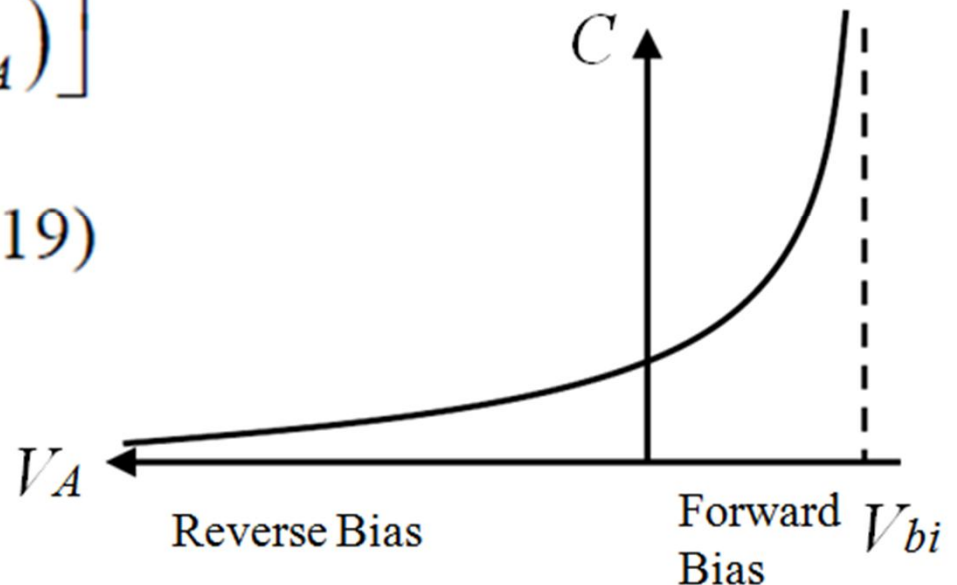
$$C_j = \frac{dQ}{dV} = \frac{d(eN_D x_n)}{dV} = eN_D \frac{d}{dV} \left\{ \sqrt{\frac{2\varepsilon}{e} \frac{N_A}{N_D N_A + N_D} (V_{bi} - V)} \right\} = \dots$$

$$C_j = \sqrt{\frac{e\varepsilon N_A N_D}{2(N_A + N_D)(V_{bi} - V)}}$$

Depletion capacitance

$$\frac{dx_n}{dV_A} = \frac{1}{N_D} \left[\frac{K_s \epsilon_o N_A N_D}{2q(N_A + N_D)(V_{bi} - V_A)} \right]^{1/2}$$

$$C' = \left[\frac{qK_s \epsilon_o N_A N_D}{2(N_A + N_D)(V_{bi} - V_A)} \right]^{1/2}$$
$$= \frac{K_s \epsilon_o}{W} \quad (19)$$



Depletion capacitance

In reverse bias, C falls as the square root of V_A . In forward bias, C approaches ∞ as V_{bi} is approached.

Actually, in forward bias, the analysis breaks down because mobile carriers in the depletion region can no longer be ignored. Equivalent expressions for C may be derived for other impurity profiles.

Measurement of C vs. V_A can be used to extract information on the doping profile.

One-sided junction

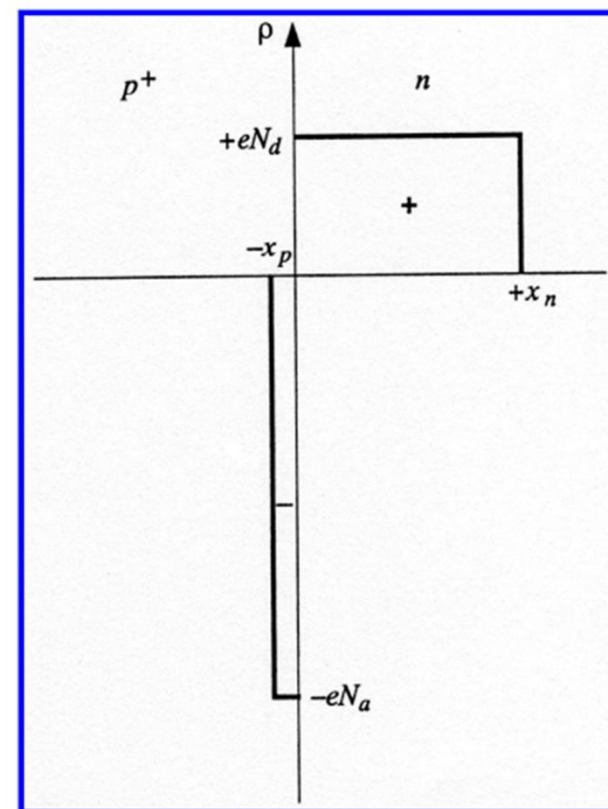
For example, $N_a \gg N_d$, a p^+n junction.

$$W = \left\{ \frac{2\epsilon_s(V_{bi} + V_R)}{e} \left[\frac{N_a + N_d}{N_a N_d} \right] \right\}^{1/2}$$

$$\approx \left\{ \frac{2\epsilon_s(V_{bi} + V_R)}{e N_d} \right\}^{1/2}$$

$$x_n = \left\{ \frac{2\epsilon_s(V_{bi} + V_R)}{e} \left[\frac{N_a}{N_d} \right] \left[\frac{1}{N_a + N_d} \right] \right\}^{1/2}$$

$$x_p = \left\{ \frac{2\epsilon_s(V_{bi} + V_R)}{e} \left[\frac{N_d}{N_a} \right] \left[\frac{1}{N_a + N_d} \right] \right\}^{1/2}$$



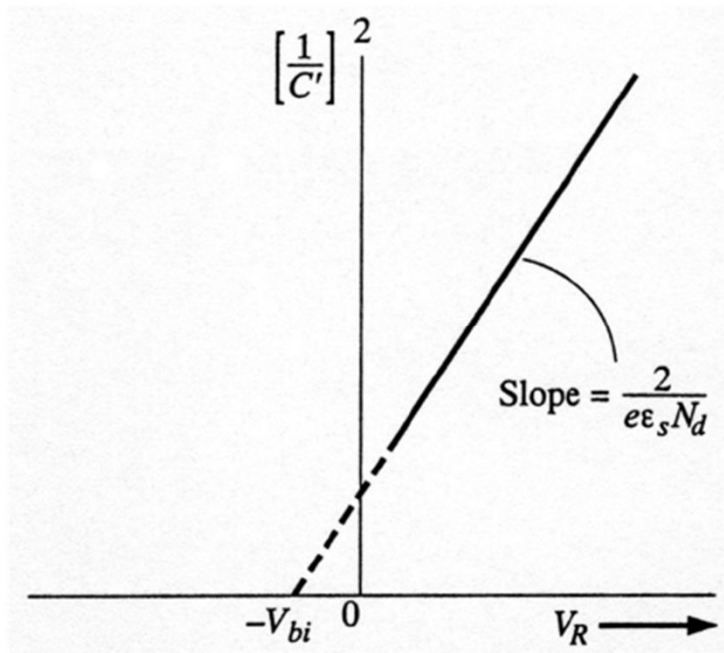
$$\Rightarrow \begin{cases} x_p \ll x_n \\ W \approx x_n \end{cases}$$

One-sided junction

For example, $N_a \gg N_d$, a p^+n junction.

$$C' = \left\{ \frac{e\epsilon_s N_a N_d}{2(V_{bi} + V_R)(N_a + N_d)} \right\}^{1/2} \approx \left\{ \frac{e\epsilon_s N_d}{2(V_{bi} + V_R)} \right\}^{1/2}$$

$$\left(\frac{1}{C'} \right)^2 = \frac{2(V_{bi} + V_R)}{e\epsilon_s N_d}$$



The left curve can be used to determine the doping concentration and the built-in potential.

One-sided junction

Example: assume a silicon p⁺n junction at $T = 300$ K with $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$. Assume that the intercept of the $(1/C')^2$ versus V_R curve gives $V_{bi} = 0.855$ volt and the slope is $1.32 \times 10^{15} [(\text{F/cm}^2)^{-2} \text{ V}^{-1}]$. Determine the doping concentrations.

$$N_d = \frac{2}{e\epsilon_s(\text{slope})} = \frac{2}{(1.6 \times 10^{-19})(11.7)(8.85 \times 10^{-14})(1.32 \times 10^{15})}$$
$$= 9.15 \times 10^{15} \text{ cm}^{-3}$$

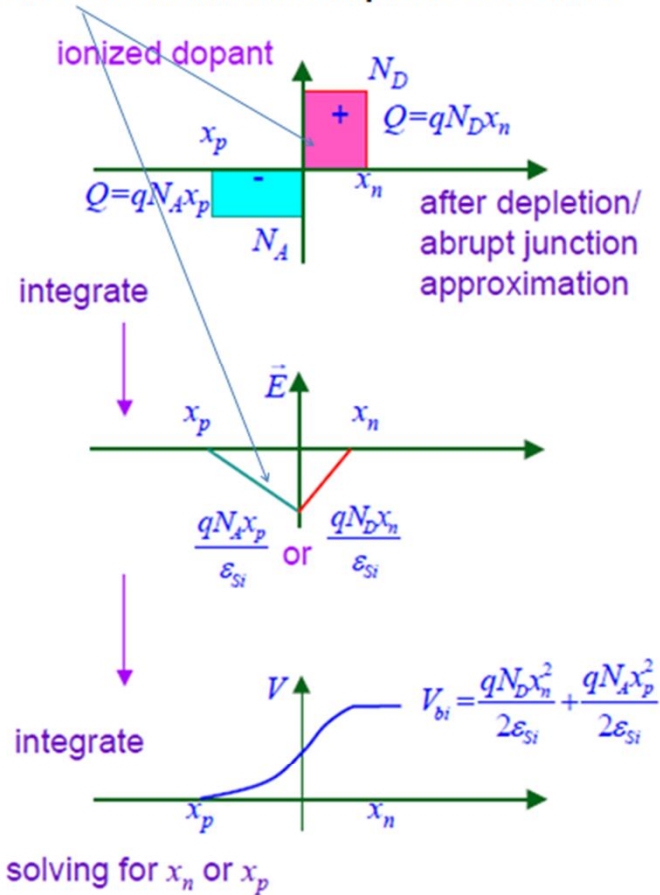
$$V_{bi} = \frac{kT}{e} \ln \left(\frac{N_a N_d}{n_i^2} \right)$$

$$N_a = \frac{n_i^2}{N_d} \exp \left(\frac{eV_{bi}}{kT} \right) = \frac{(1.5 \times 10^{10})^2}{9.15 \times 10^{15}} \exp \left(\frac{0.855}{0.0259} \right)$$

$$N_a = 5.34 \times 10^{18} \text{ cm}^{-3}$$

Summary

The area determines the amplitude of E and V



$$1. \quad V_{bi} = \frac{kT}{q} \ln \frac{N_D N_A}{n_i^2}$$

$$2. \quad x_n = \sqrt{\frac{2\epsilon}{e} \frac{N_A}{N_D N_A + N_D^2} V_{bi}}$$

$$x_p = \sqrt{\frac{2\epsilon}{e} \frac{N_D}{N_D N_A + N_A^2} V_{bi}}$$

$$x_d = \sqrt{\frac{2\epsilon}{e} \frac{(N_A + N_D)}{N_D N_A} V_{bi}}$$

$$3. \quad E(x) = -\frac{eN_A}{\epsilon} (x + x_p)$$

$$E(x) = \frac{eN_D}{\epsilon} (x - x_n)$$

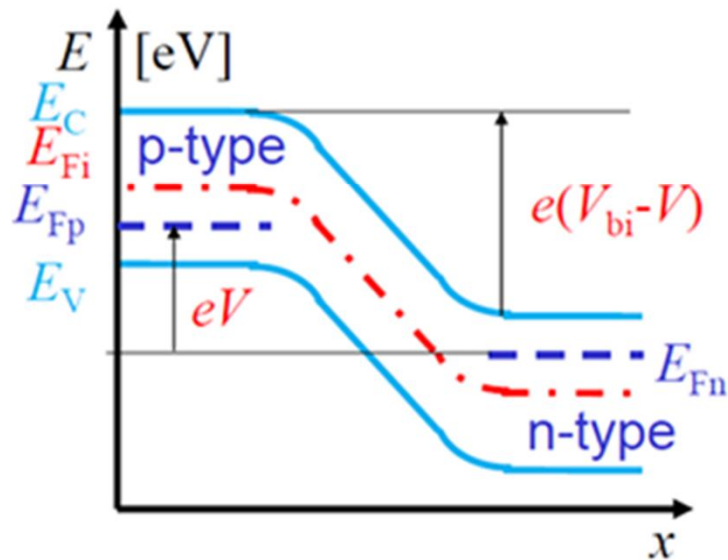
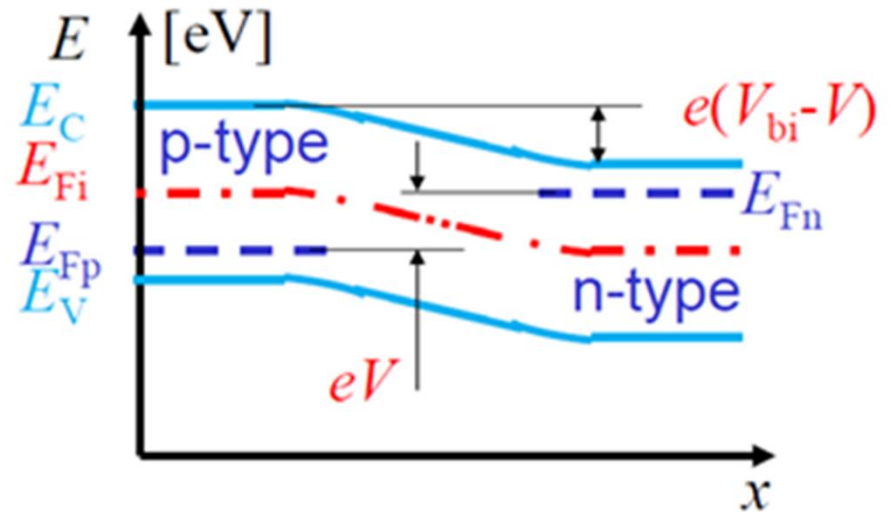
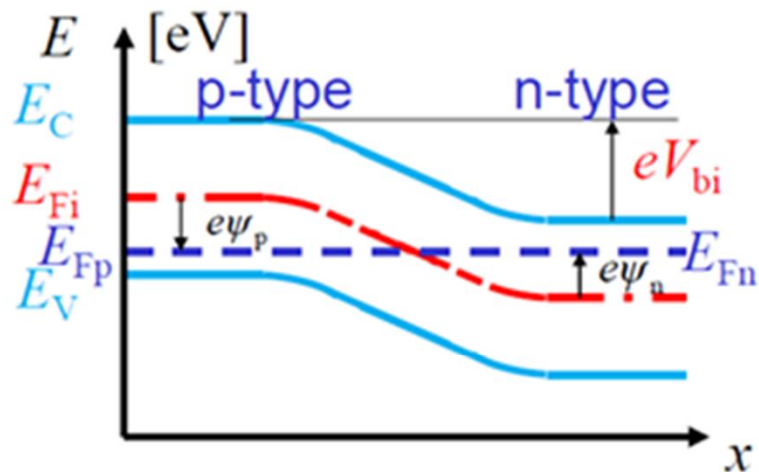
$$\psi(x) = \frac{eN_A}{2\epsilon} (x_p + x)^2$$

4.

$$\psi(x) = V_{bi} - \frac{eN_D}{2\epsilon} (x_n - x)^2$$

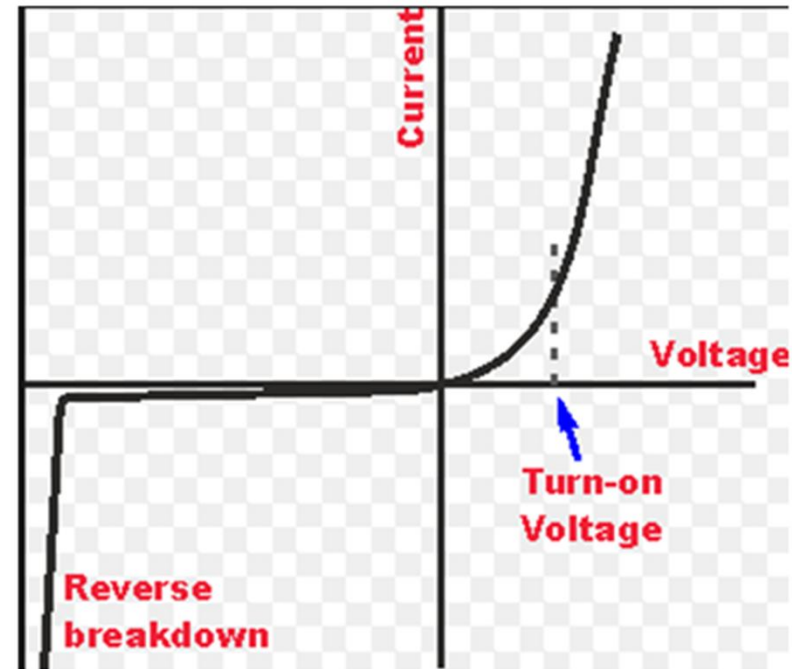
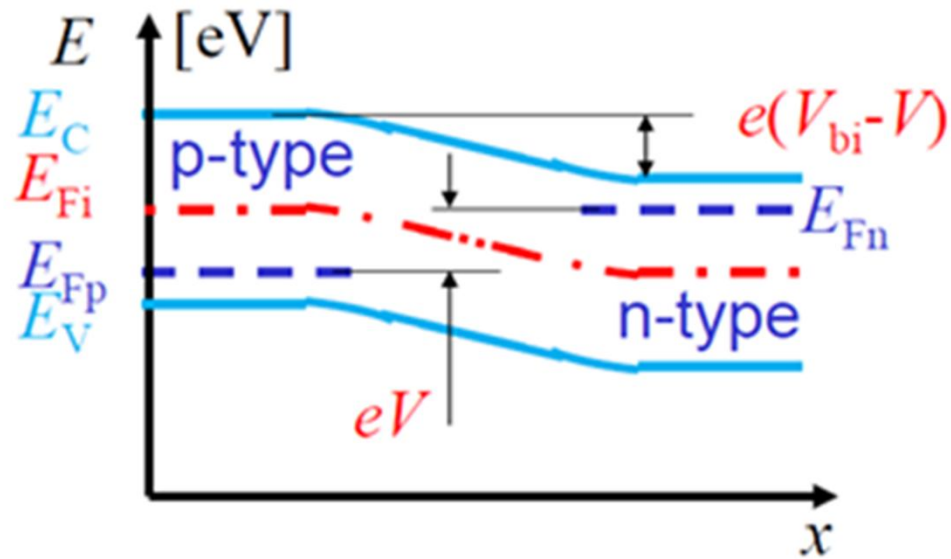
$$5. \quad C = \frac{K_s \epsilon_o}{W}$$

Quiz: what bias was applied?

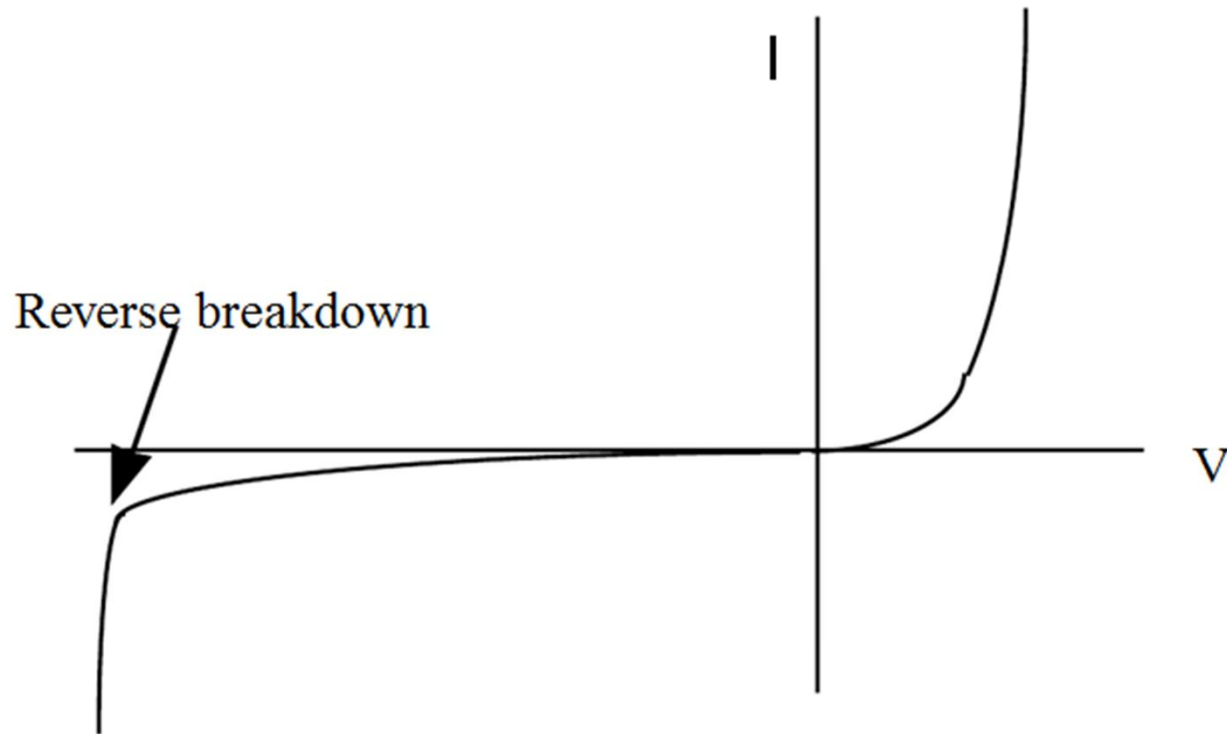


- Quiz: width of depletion increases or decreases under:
1. Forward bias
 2. Reverse bias
 3. Increase Na or Nd

Quiz: what happen if external bias is larger than V_{bi} ?



7.4 | JUNCTION BREAKDOWN

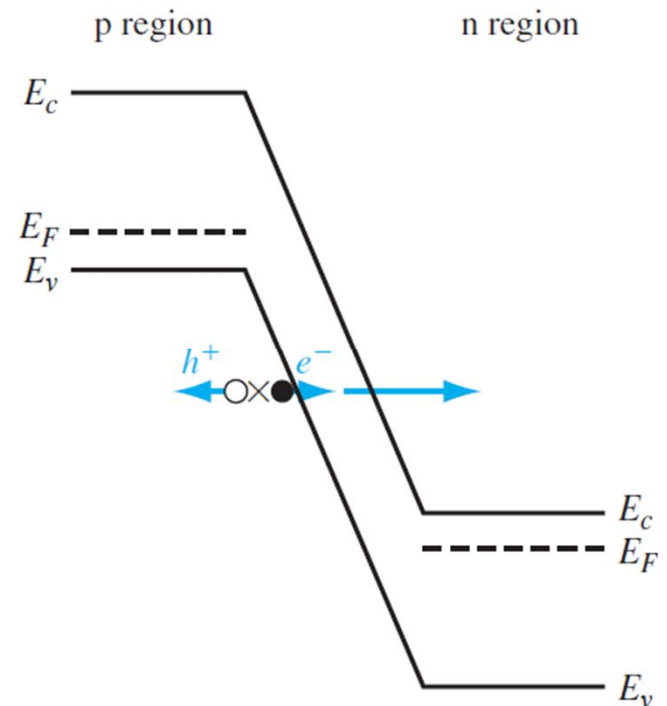


Breakdown is not explained at all by the ideal diode equations, hence represents an entirely different phenomena.

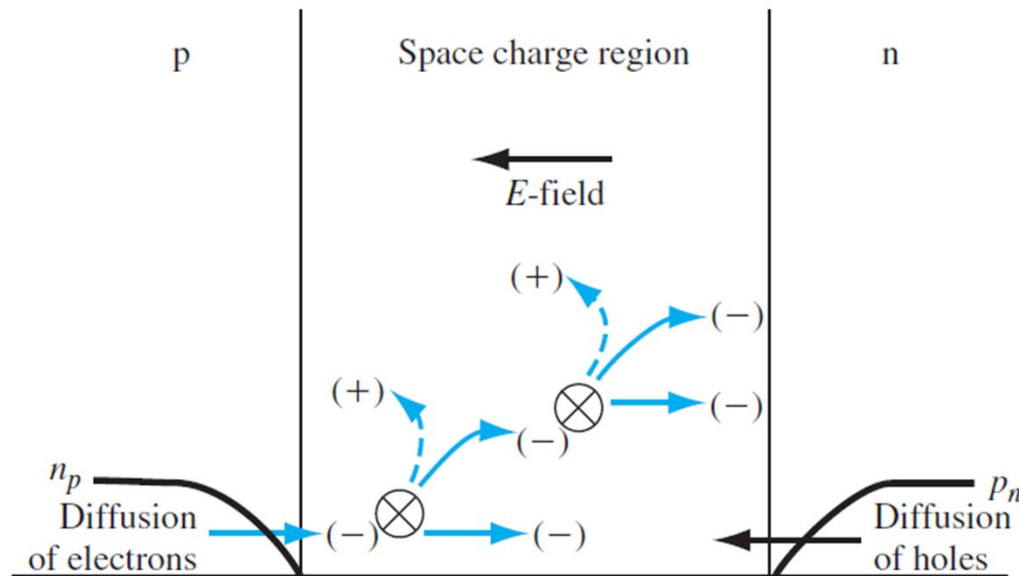
7.4 | JUNCTION BREAKDOWN

The reverse-bias voltage applied on pn junctions cannot increase without limit. Up to a certain threshold, the reverse-bias current will increase rapidly. The applied voltage at this point is called the **breakdown voltage**. There are two mechanisms causing the reverse-bias breakdown in a pn junction: the **Zener** effect and the **avalanche** effect.

Zener breakdown occurs in highly doped pn junctions through a tunneling mechanism. The conduction and valence bands on the opposite sides of highly doped pn junctions are very close under reverse bias so that electrons may tunnel directly from the valence band on the p-side into the conduction band on the n-side.

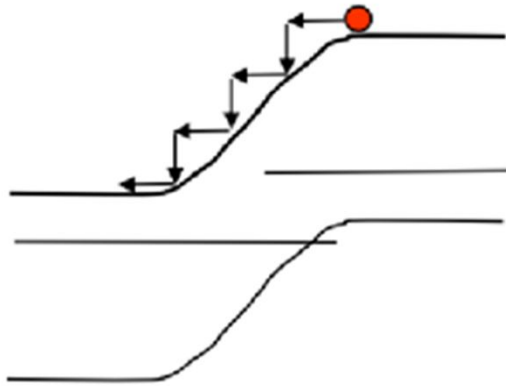


Avalanche Breakdown

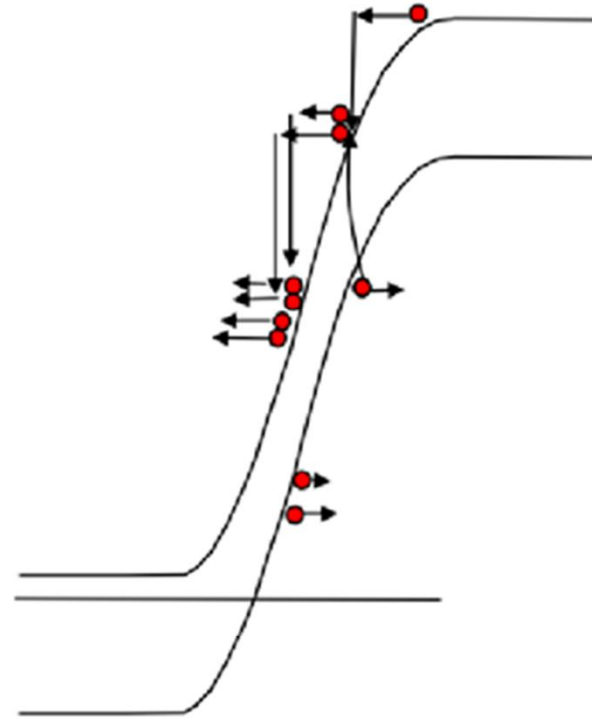


Avalanche breakdown occurs when electrons/holes moving through the space charge region acquire sufficient energy from the electric field to create electron-hole pairs by colliding with atoms. The newly created electrons/holes move in opposite directions due to the electric field and thus contribute to the reverse-bias current. In addition, the newly generated electrons/holes may acquire sufficient energy to ionize other atoms, leading to the avalanche process.

Avalanche Breakdown



(a) Small reverse bias



(b) Large Reverse bias $\rightarrow V_{BD}$

Avalanche Breakdown

Electrons entering the depletion region (or generated there by R-G centers), gain kinetic energy under the influence of the electric field, E and subsequently lose it as heat through scattering. If E is high enough, the electron can gain sufficient energy between scattering events to break covalent bonds, freeing an electron and creating a hole. This process is repeated resulting in large reverse currents. Since the carrier must gain an energy, $E \geq E_g$, a critical field is required to initiate avalanche breakdown since at lower fields, carriers will scatter without causing impact ionization.

Electron and Hole Currents in Avalanche

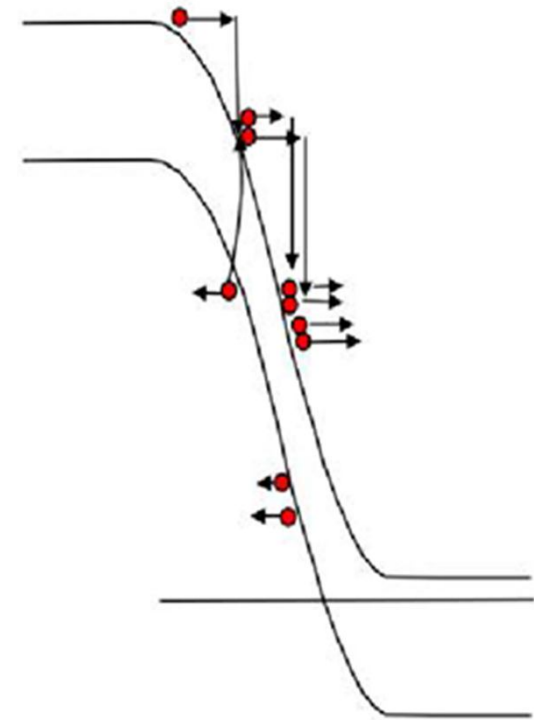
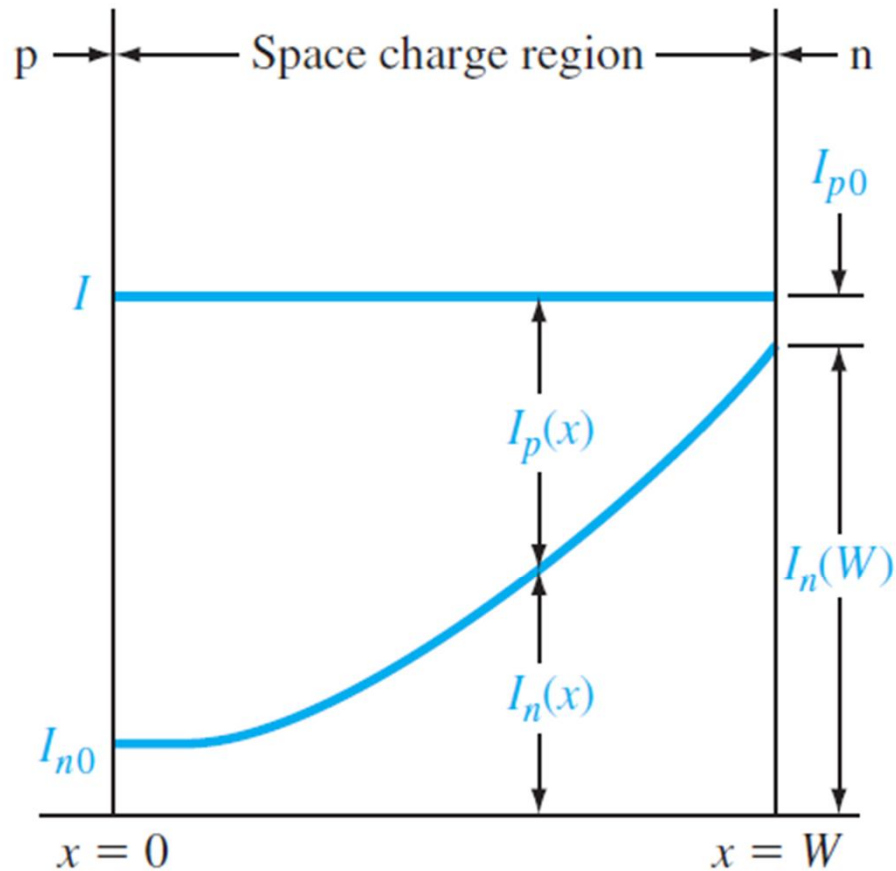
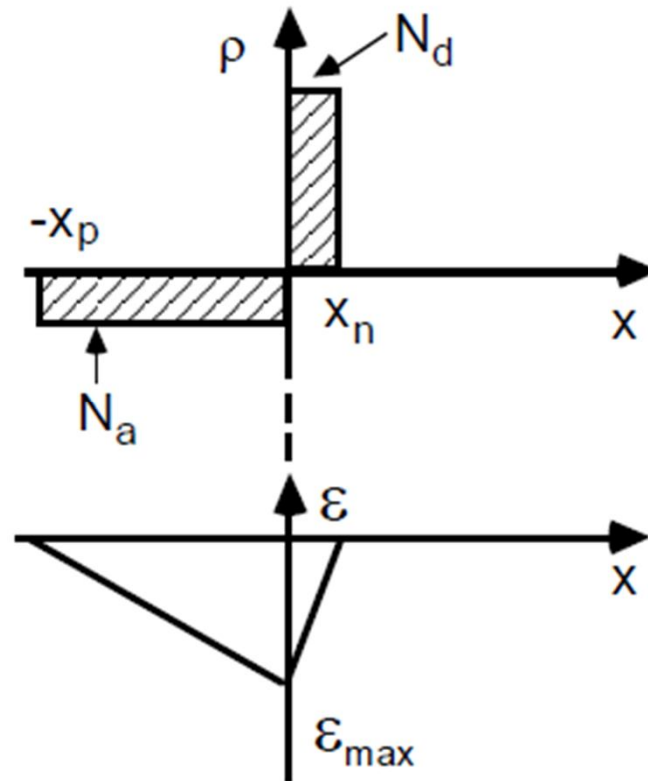


Figure 7.13 | Electron and hole current components through the space charge region during avalanche multiplication.

7.4 | JUNCTION BREAKDOWN



Most of the depletion region occurs on the lightly doped side.

$$W \approx x_p = \left[\frac{2K_s \epsilon_o}{qN_A} (V_{bi} - V_A) \right]^{1/2}$$

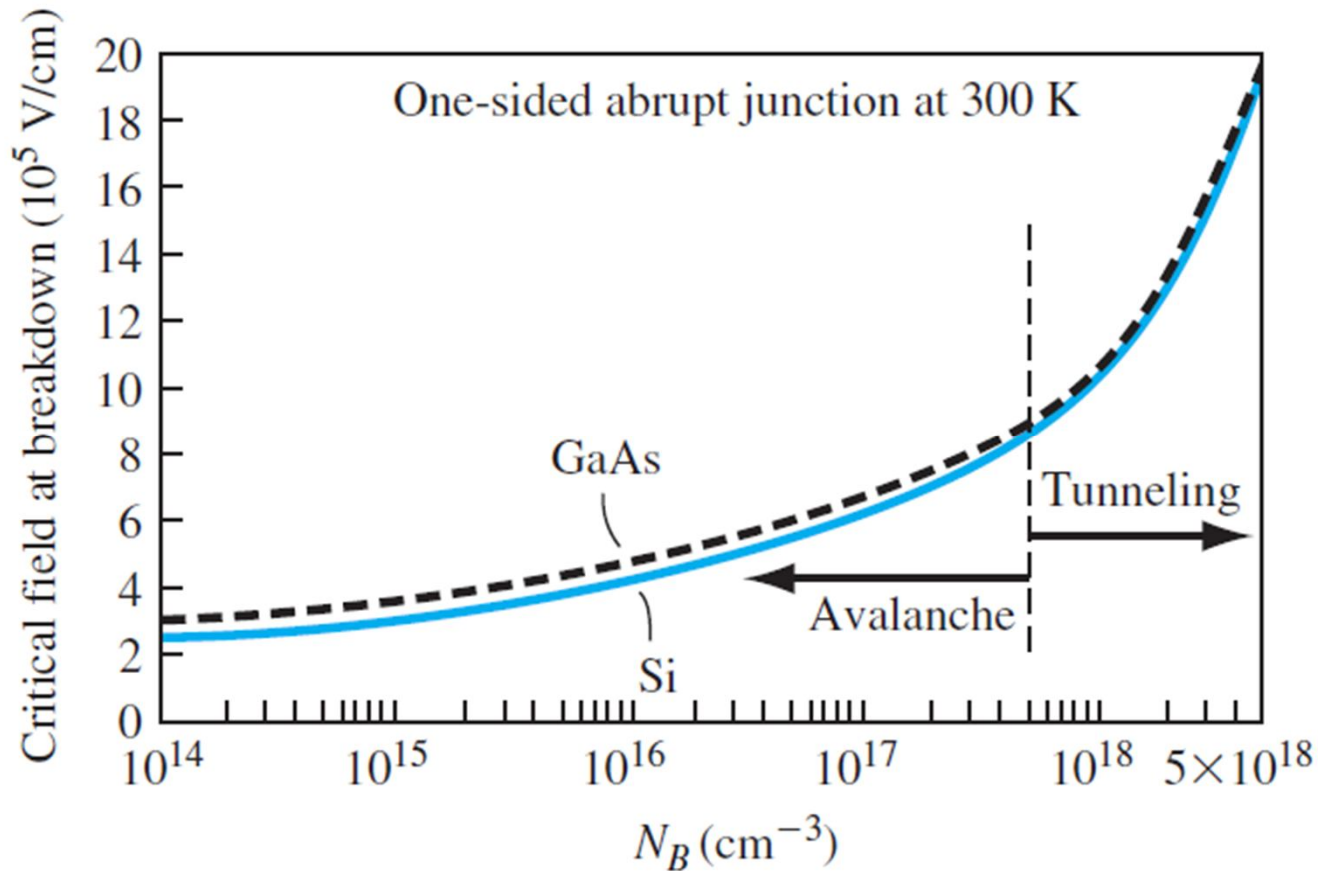
7.4 | JUNCTION BREAKDOWN

$$E_{max} = \frac{eN_A x_p}{K_S \epsilon_0} = \left(\frac{2eN_A(V_{bi} - V_A)}{K_S \epsilon_0} \right)^{1/2}$$

If breakdown is defined to occur when the maximum field reaches the critical field $\mathcal{E}_{CR}$, then

$$V_{BR} \approx \frac{K_s \epsilon_o \mathcal{E}_{CR}^2}{2qN_A}$$

Critical breakdown fields in Si and GaAs



Note that the breakdown field is determined by the doping on the lightly doped side.

Critical breakdown fields in Si and GaAs

Note that the critical electric field, $\mathcal{E}_{CR}$, for avalanche breakdown increases moderately as N_D or N_A increases. This is because ionized impurity scattering increases and on average, slows down carriers by scattering more frequently.

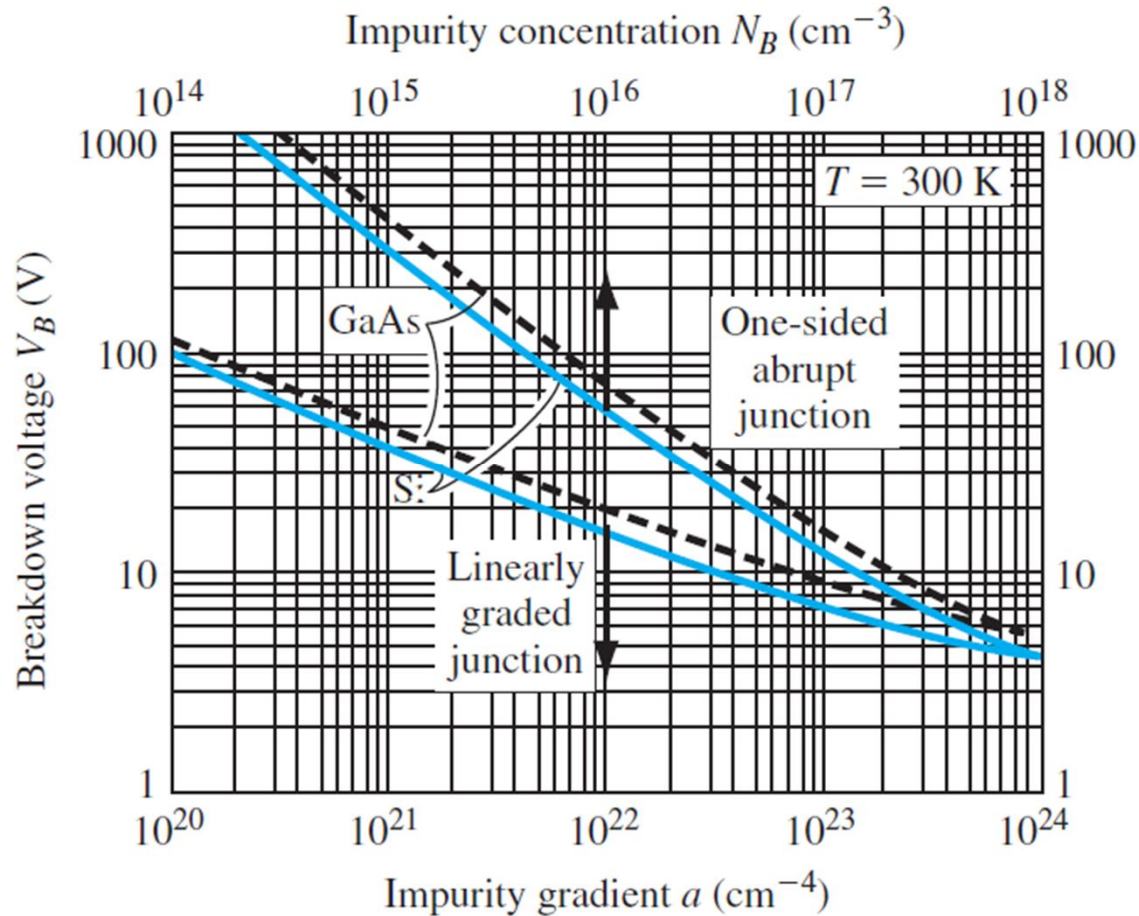
Since the critical field is $\approx$ constant at $\mathcal{E}_{CR} \approx 3 \times 10^5 - 8 \times 10^5$ V/cm, the breakdown voltage varies approximately inversely as the doping on the lightly doped side of the junction.

$$V = uE = 1400 * 10^5 = \sim 10^8 \text{ cm/s} = 10 * V_{\text{thermal}}$$

$$\text{Energy} = 100 * \text{thermal energy} = 3.9 \text{ eV} > 1.1 \text{ eV}$$

PN Junction Breakdown Voltage

$$V_{BR} \approx \frac{K_s \epsilon_o \mathcal{E}_{CR}^2}{2qN_A}$$



Quiz: GaAs vs. Si which has higher V_{BR} ?

Breakdown voltage

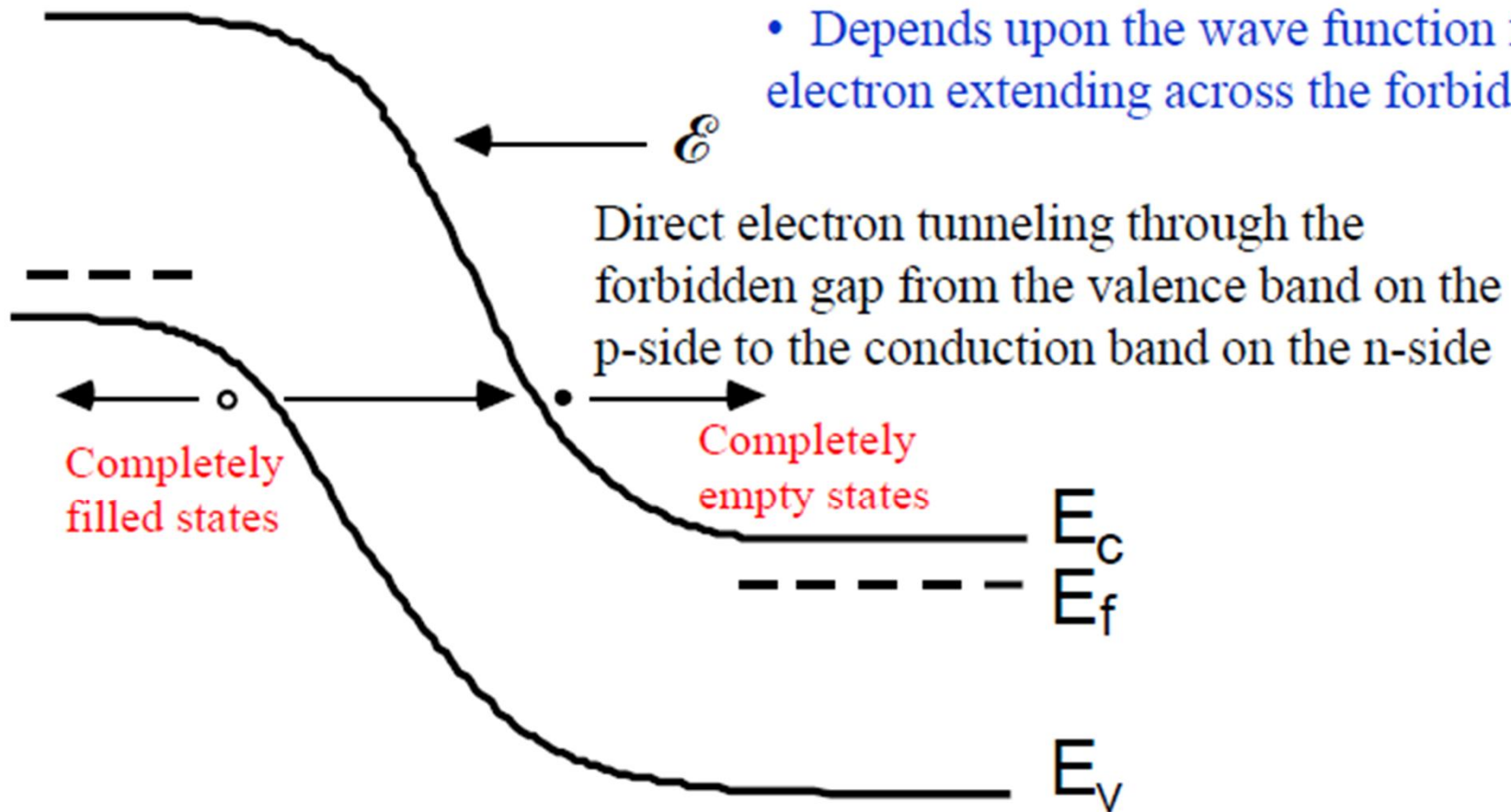
Note:

1. For breakdown voltage $>$ a few volts, the inverse relationship of V_{BR} to N_A or N_D holds reasonably well.
2. At high doping levels, the V_{BR} flattens out because a different mechanism (Zener **breakdown or tunneling**) becomes dominant.
3. V_{BR} increases at a given N_A or N_D as the bandgap of the material increases. This makes sense because it is harder to create electron hole pairs in wider bandgap material, increasing $\mathcal{E}_{CR}$.
4. Similar calculations may be carried out for linearly graded junctions with generally similar results. Higher gradient results in lower V_{BR} .

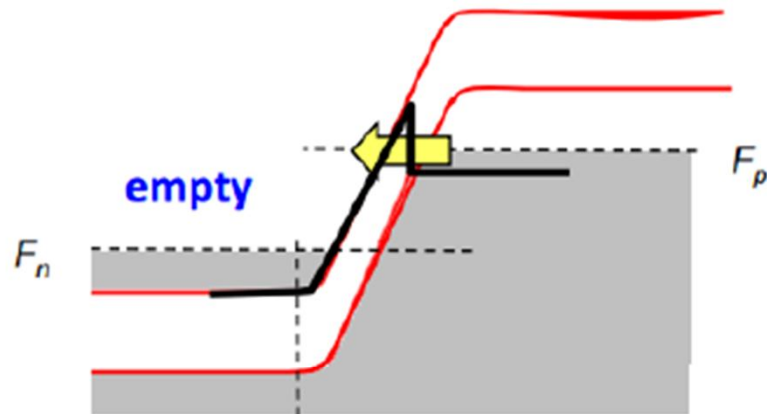
Tunneling or Zener Breakdown

Tunneling is a quantum mechanical phenomena

- Depends upon the wave function for an electron extending across the forbidden gap



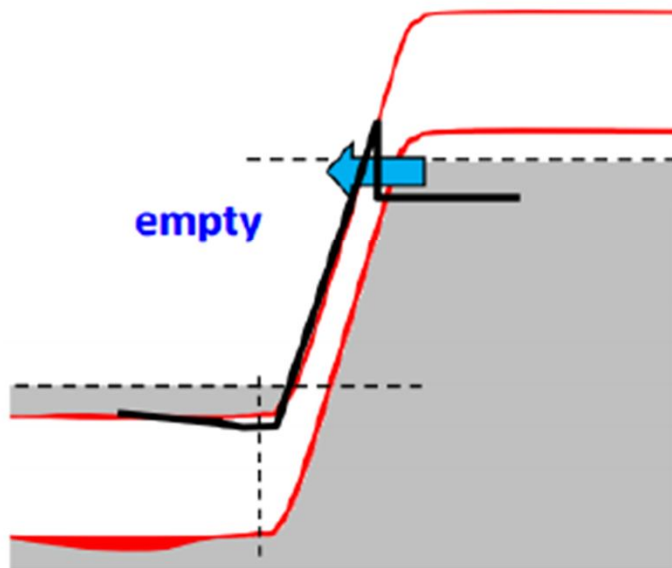
Tunneling or Zener Breakdown



High doping $\rightarrow$ high E,
less W_D

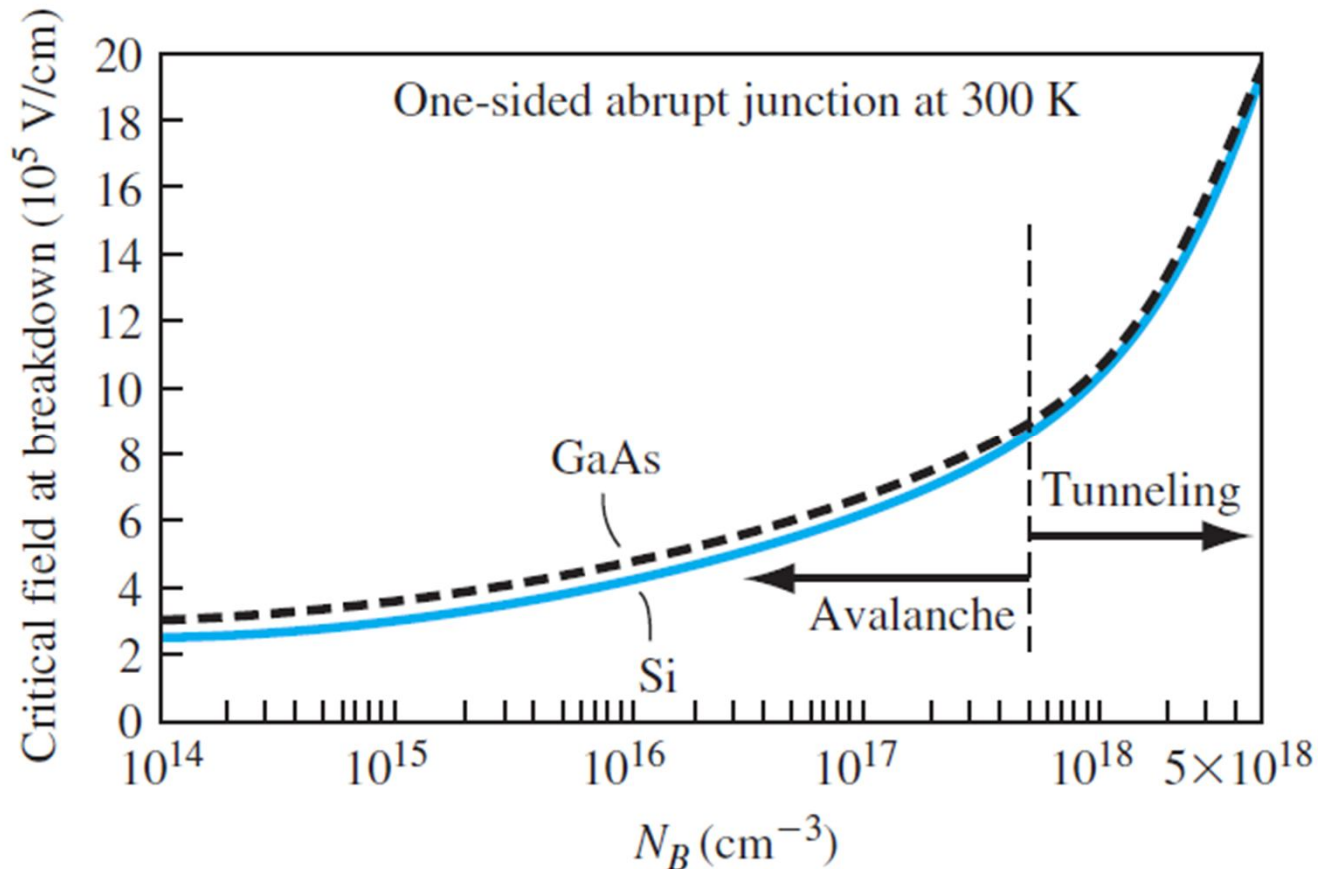
$$x_d = \sqrt{\frac{2\epsilon}{e} \frac{(N_A + N_D)}{N_D N_A} V_{bi}}$$

High doping necessary
for Zener Breakdown
or Tunneling



*This figure extreme
case $\rightarrow$ degenerate
doping*

Critical breakdown fields in Si and GaAs



Note that the breakdown field is determined by the doping on the lightly doped side.

Tunneling or Zener Breakdown

If the field is high enough ($\mathcal{E} > 10^6$ V/cm) a Si-Si bond can be broken in scattering, creating an electron hole pair directly.

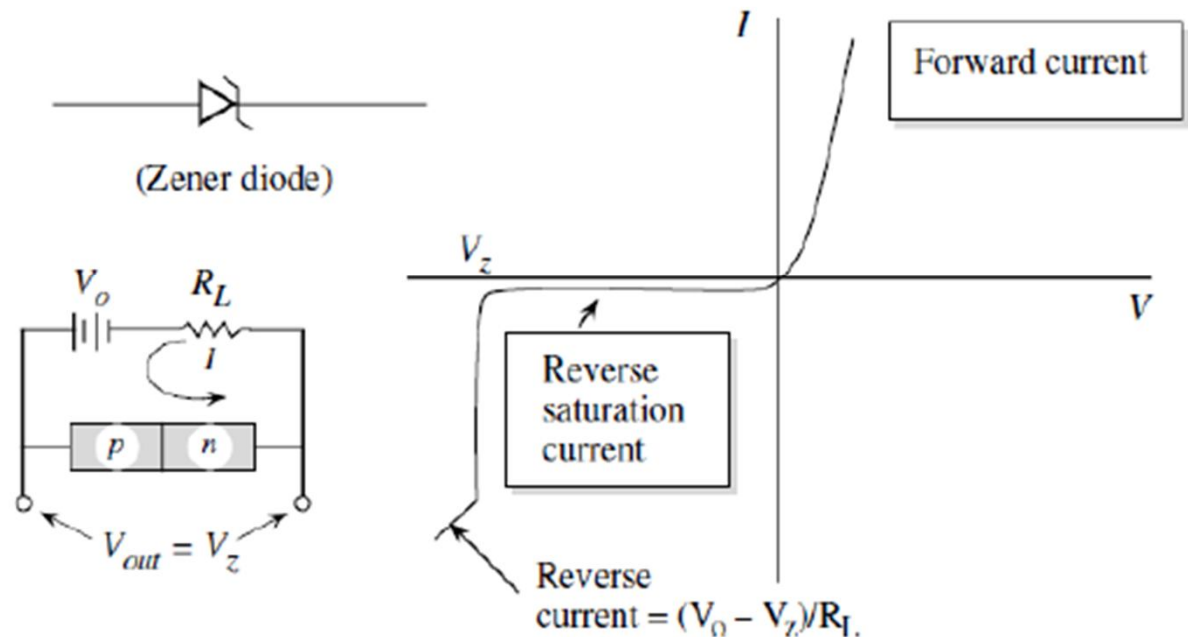
Alternatively, an electron from the valence band can tunnel into the conduction band. This process is likely only when the doping levels on both sides of the junction are very high ($> 10^{18}$ cm⁻³) so that W is very small (< 100 Å).

In general, if a junction shows a breakdown voltage $< 4E_g/q$, then Zener breakdown is occurring. For breakdown voltage $> 6E_g/q$, avalanche breakdown dominates. Also, if the breakdown is independent of temperature, it is due to tunneling and is Zener breakdown.

Tunneling or Zener Breakdown

There is a special class of diodes called **Zener diodes** where tunneling is exploited. The depletion width can be controlled by the doping density. If the junction is made from heavily doped materials, the Zener tunneling can start at a reverse bias of V_z , which could be as low as a few tenths of a volt. The voltage across the junction is then clamped at V_z , and the current is controlled by the external circuit as shown in the figure. This clamping property provides a very useful application for the Zener diodes. If V_z is breakdown voltage (due to impact ionization or Zener breakdown), the current for reverse bias voltages greater than V_z is

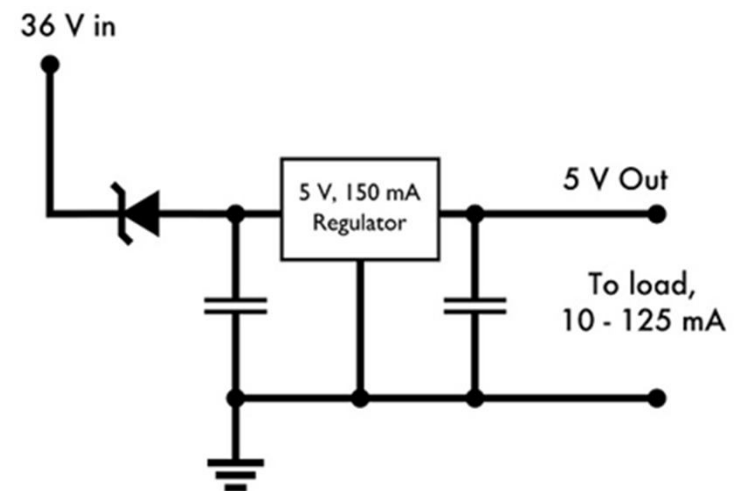
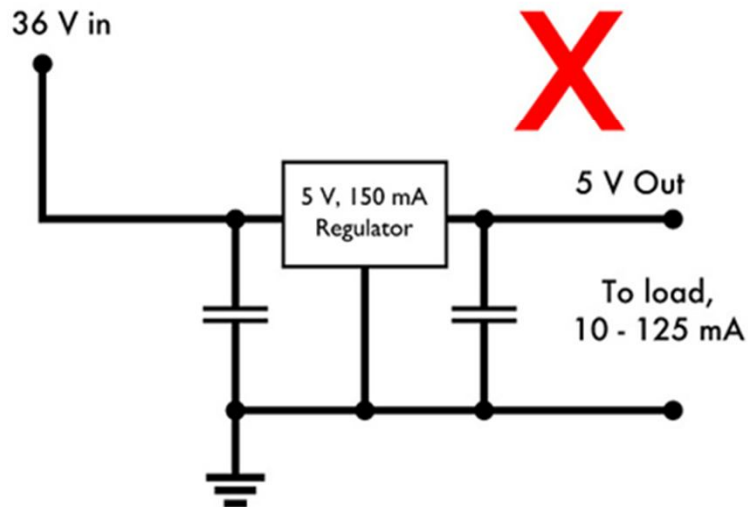
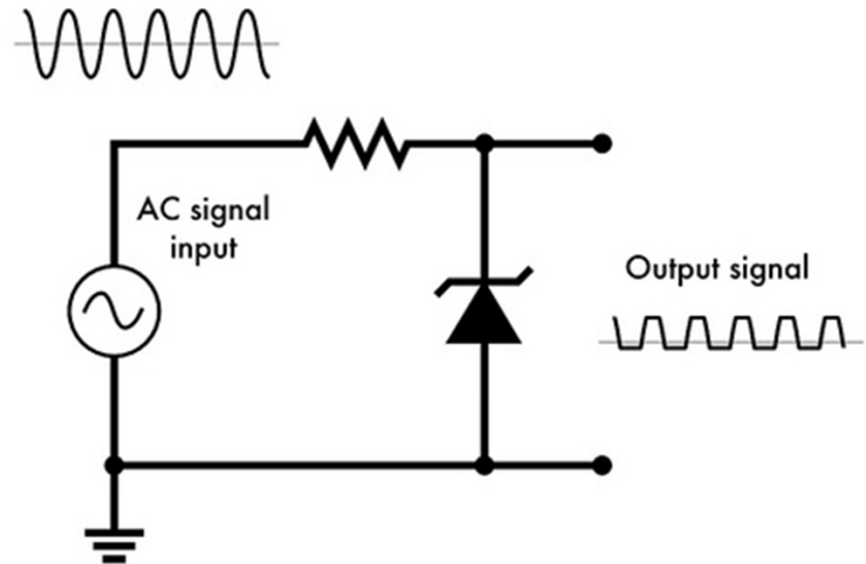
$$I = \frac{|V - V_z|}{R_L}$$



输出电流可以很大，但输出电压保持不变

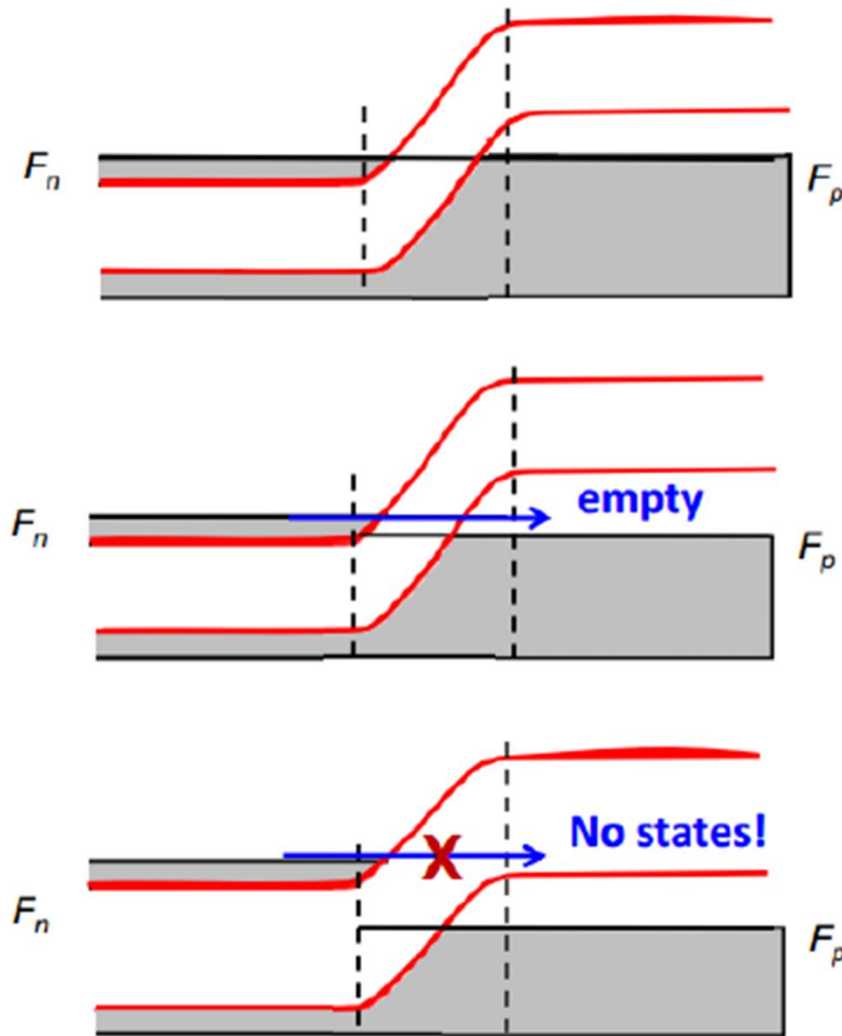
Tunneling or Zener Breakdown

Zener diode applications

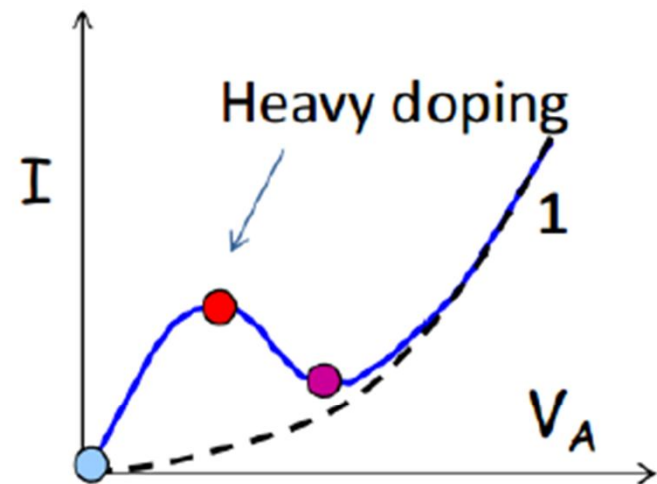


Tunneling or Zener Breakdown

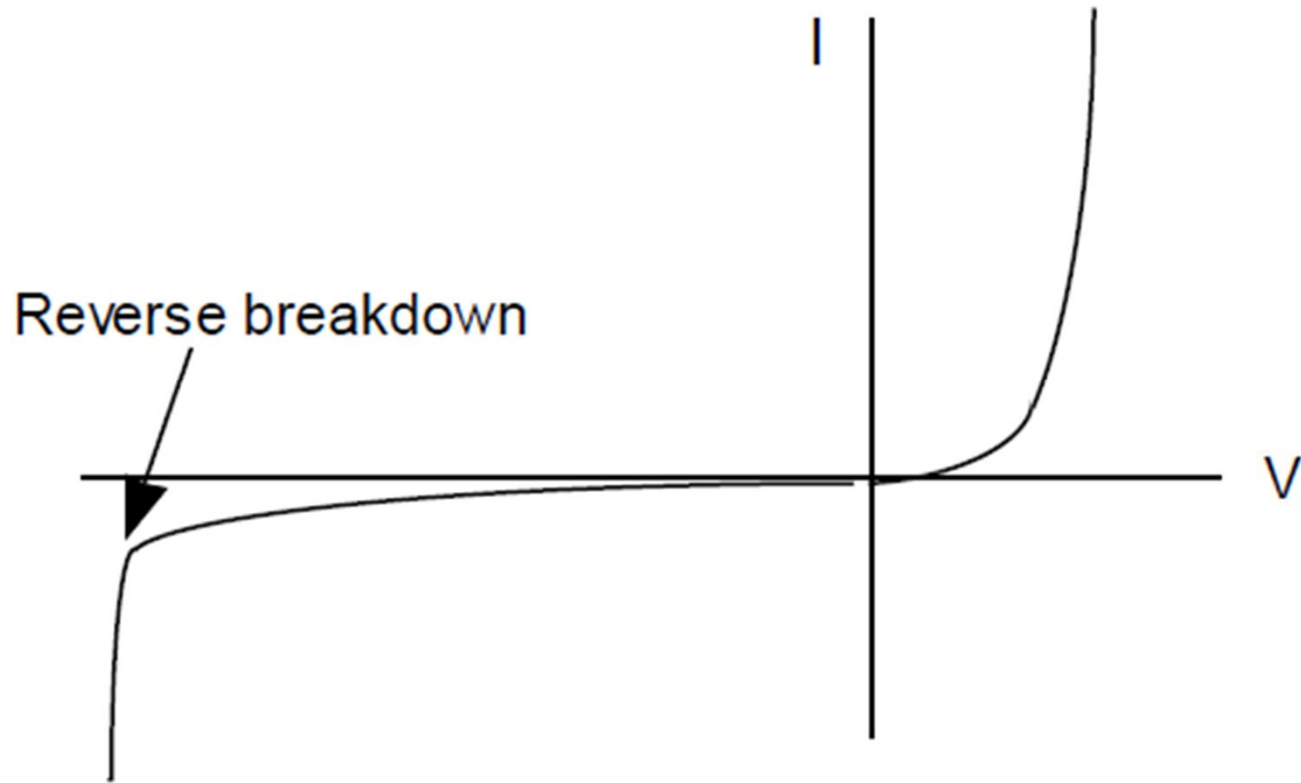
Tunneling in forward bias?: Esaki diode



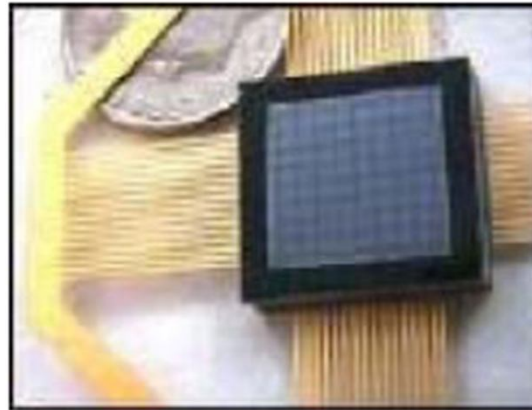
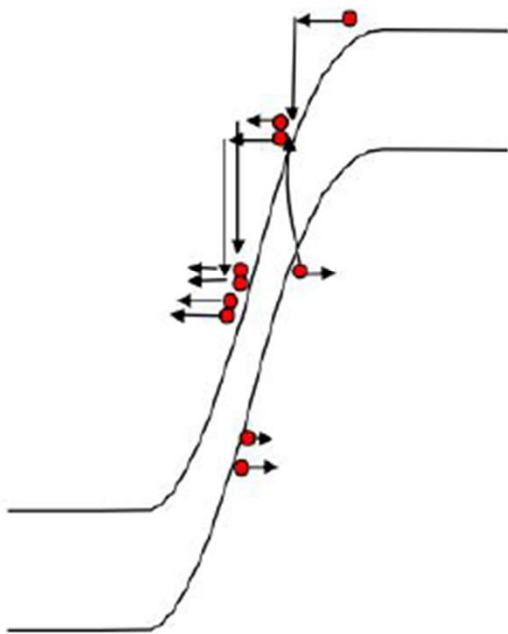
Discovered by
Leo-Esaki in 1958



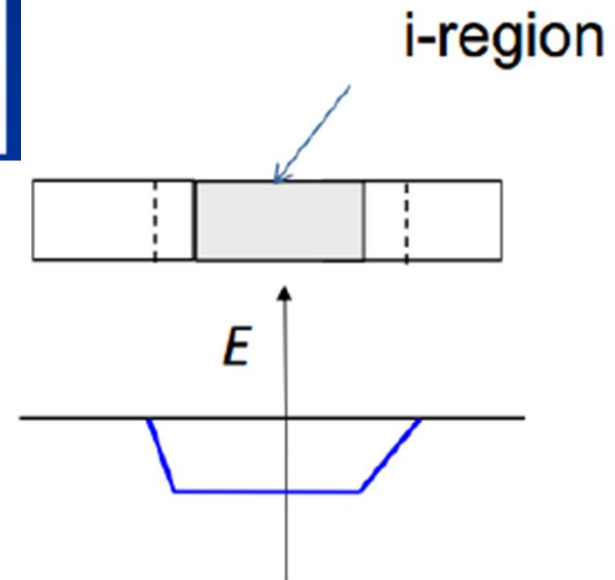
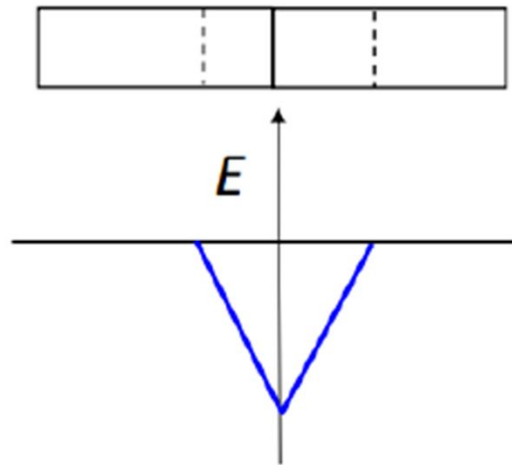
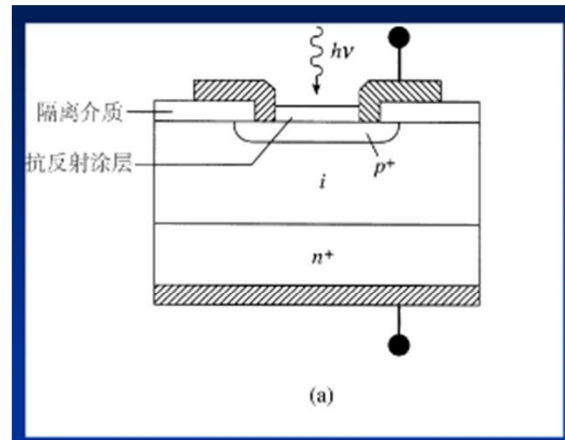
Is diode breakdown permanent?



Avalanche Photodiode (APD)



Junction Engineering



Reduced field for p-i-n junction, because V_{bi} (area under the curve) must be the same.

And therefore we should increase the V_{br} for the p-i-n junction so that avalanche process can occur.

Reading Materials

D. A. Neamen, “Semiconductor Physics and Devices: Basic Principles”, Fourth edition, Chapter 7, “The pn Junction”.